



Comparing the incidence of bone tumors in rats chronically exposed to the selective PTH type 1 receptor agonist abaloparatide or PTH(1–34)

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ABSTRACT

Prolonged treatment with human parathyroid hormone (hPTH) in rats results in development of bone tumors, though this finding has not been supported by clinical experience. The PTH type 1 receptor agonist abaloparatide, selected for its bone anabolic activity, is under clinical development to treat postmenopausal women with osteoporosis. To determine the carcinogenic potential of abaloparatide, Fischer (F344) rats were administered SC daily abaloparatide at doses of 0, 10, 25, and 50 µg/kg or 30 µg/kg hPTH(1–34) as a positive control for up to 2 years. Robust increases in bone density were achieved at all abaloparatide doses and with hPTH(1–34). Comprehensive histopathological analysis reflected a comparable continuum of proliferative changes in bone, mostly osteosarcoma, in both abaloparatide and hPTH(1–34) treated rats. Comparing the effects of abaloparatide and hPTH(1–34) at the 25 and 30 µg/kg respective doses, representing similar exposure multiples to the human therapeutic doses, revealed similar osteosarcoma-associated mortality, tumor incidence, age at first occurrence, and metastatic potential. There were no increases in the incidence of non-bone tumors with abaloparatide compared to vehicle. Thus, near life-long treatment with abaloparatide in rats resulted in dose and time dependent formation of osteosarcomas, with a comparable response to hPTH(1–34) at similar exposure.

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1. Introduction

Abaloparatide is a synthetic agonist of the parathyroid hormone type 1 receptor (PTH1R) and bone anabolic agent that is currently under development for the treatment of postmenopausal osteoporosis (PMO). When injected subcutaneously (SC), abaloparatide increases bone formation with lesser effects on bone resorption, resulting in increased bone mass in animals (Bahar et al., 2016) and in humans (Leder et al., 2015). Recent evidence has demonstrated that abaloparatide reduced both vertebral and nonvertebral fractures in PMO patients after 18 months of daily SC administration (Miller et al., 2016).

Abaloparatide was originally selected for its bone anabolic

activity and more limited effect on bone resorption and calcium mobilization relative to hPTH(1–34). With 41% homology to PTH(1–34) and 76% homology to PTHrP(1–34), its greater homology to PTHrP reflects the ability of PTHrP to increase bone formation with less stimulation of bone resorption and hypercalcemia (Horwitz et al., 2003). These differential actions are mediated by distinct PTH1R conformation selectivity, with PTHrP binding with greater selectivity to the RG conformation (relative to R0) than PTH (Hattersley et al., 2016). Efficient binding to R0 results in prolonged signaling responses in cells and prolonged calcemic responses, whereas selective binding to RG results in more transient responses (Fevrandon et al., 2009). Abaloparatide binds with greater selectivity to the RG conformation than does hPTH(1–34) or hPTHrP, and as a result of this RG bias, abaloparatide mediates more transient cAMP responses in PTH1R-expressing cells (Hattersley et al., 2016). These findings provide a mechanistic basis – i.e., transient intracellular signaling via RG-selective binding – that can account for

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estimation. All parameters were generated from abaloparatide concentrations in plasma from Week 1, 26, and 52. Parameters were estimated using sampling times relative to the start of each dose administration.

The presence of anti-abaloparatide antibodies were assessed in serum samples collected prior to treatment and prior to termination for TK animals (week 52) and main study animals by radioimmunoassay at PPD Inc. Anti-drug antibodies were not assessed in the hPTH(1–34) positive control group.

2.3. Statistics

All statistical analyses were performed using SAS (version 8.2; SAS Institute, Cary, NC). The survival function of each group was estimated using the Kaplan-Meier product-limit method applied on daily intervals. For the males' and females' datasets, the significance of group effects on mortality rates was determined among the Vehicle and abaloparatide groups. Group homogeneity was first determined by a Logrank test, with significant heterogeneity resulting in implementation of a Peto's two-sided test and examination of a dose-response relationship. Significance was determined based on $p \leq 0.05$.

For the incidence tumor data, neoplastic lesions categorized as palpable were analyzed in a mortality-independent context according to Peto's onset rate method (Peto et al., 1980). Tumors categorized as nonpalpable were analyzed using the mortality-dependent method. In this context, each neoplastic finding was classified as the cause of death (fatal) or not the cause of death (incidental) and was analyzed accordingly using Peto's death rate and the prevalence methods, as per the US FDA's Guidance for Industry (FDA, 2001). The processing of incidental tumors was done by creating a single separate interval for the time period following the experimental period (terminal sacrifice period) and by dividing the experimental period into fixed intervals.

For the DXA data, the vehicle and abaloparatide groups were analyzed by one way analysis of variance (ANOVA). If the group variances were homogenous ($p > 0.05$ by Levene's test), group differences from vehicle controls were determined by Dunnett's post-test, otherwise they were determined by Dunn's test. Differences between the vehicle and hPTH(1–34) group were determined by two-sample *t*-test, or a Wilcoxon-rank-sum test if group variances were significantly different ($p > 0.05$ by Levene's test).

3. Results

3.1. Drug exposure

Peak abaloparatide concentrations were generally observed between 0.25 and 0.5 h postdose, with half-life ($t_{1/2}$) estimated to

be between 0.221 and 0.631 h. There were no noteworthy differences in exposure between genders, thus the data were combined for modeling purposes. Exposure generally increased with dose level, with greater than dose proportionally at Weeks 26 and 52 when some dose accumulation was evident (Table 1). The week 52 toxicokinetic data from this carcinogenicity study and pharmacokinetic data from a human Phase I study after 7 days of dosing was used to determine the relative differences in abaloparatide exposure. Based on the AUC data from female rats at week 52 and from postmenopausal women administered abaloparatide at 80 µg SC once daily for 7 days, the exposure multiples in this study were approximately 4-, 16- and 28-fold at the 10, 25 and 50 µg/kg dose levels respectively. Anti-abaloparatide antibodies were seen in one female at 25 µg/kg and 2 females at 50 µg/kg at Week 52 (TK) and in 2 main study females at 50 µg/kg prior to termination.

3.2. Mortality

Male rats treated with abaloparatide at ≥ 25 µg/kg and hPTH(1–34) at 30 µg/kg had a significant increase in mortality rate relative to Vehicle and were terminated early at weeks 97, 88 and 89, respectively when the number of survivors reached 15/group (Table 2). All female groups were euthanized at week 104, though a significant increase in mortality rate was observed in the 50 µg/kg abaloparatide group. These differences in mortality rate were attributed to osteosarcomas, which were more prevalent in males than females irrespective of the compound administered. Notably, mortalities secondary to osteosarcoma did not differ in rats treated with 25 µg/kg abaloparatide when compared to rats treated with 30 µg/kg hPTH(1–34). Fatal osteosarcomas were first observed in abaloparatide-treated male rats on Day 310 and in female rats on Day 398, and with hPTH(1–34) on Days 343 and 419 in male and females respectively.

3.3. Bone radiodensity and mass

Increased radio-opacity was observed throughout the skeleton in nearly all animals treated with abaloparatide and hPTH(1–34), as illustrated for abaloparatide in Fig. 1. These changes corresponded to significant increases in bone mineral content (BMC) by DXA at the lumbar spine and whole femur (Fig. 1). Lumbar spine BMC was increased +61, +66, and 80% relative to vehicle controls for males at 10, 25, and 50 µg/kg, respectively, and +58, +92, and 93% for females. Increases in BMC in the 25 and 50 µg/kg abaloparatide groups were generally comparable to the changes in the 30 µg/kg hPTH(1–34) group. At the femur, the abaloparatide- and hPTH(1–34)-mediated changes in whole femur BMC were due to similar increases at both cortical (femur diaphysis) and mixed trabecular/cortical (distal femur) sites.

Abbreviations	
ABL	abaloparatide
ANOVA	analysis of variance
AUC	area-under-the-curve
BMC	bone mineral content
CHMP	Committee for Medicinal Products for Human Use
DXA	dual energy x-ray absorptiometry
F344	Fischer 344 (rat)
FDA	Food and Drug Administration
FOH	Focal osteoblast hyperplasia
HaE	Hematoxylin and eosin (stain)
hPTH	human PTH (segment 1–34 or full length 1–84 specified)
LGL	large granular lymphocytic (leukemia)
PMO	postmenopausal osteoporosis
PPD Inc	Pharmaceutical Product Development Inc.
PTH	parathyroid hormone
PTH1R	PTH type 1 receptor
PTHrP	PTH-related Peptide
rhPTH	recombinant human PTH (segment 1–34 or full length 1–84 specified); SC = subcutaneous; TK = toxicokinetic

the favorable anabolic effects that abaloparatide has demonstrated in preclinical and clinical studies.

Two recombinant human PTH peptides have been approved for the treatment of PMO. Teriparatide, or rhPTH(1–34), is the first 34 amino acids of PTH and has been marketed at a 20 µg SC daily dose for the treatment of PMO in the United States since 2002 and in Europe since 2003. The full-length rhPTH(1–84) was approved in Europe since 2006 at a daily dose of 100 µg SC. Based on their stimulatory effects on osteoblasts, chronic 2-year studies were performed in rats to determine their potential impact on carcinogenicity. These studies demonstrated dose and duration-related osteosarcomas with rhPTH(1–34) (Vahle et al., 2002, 2004) and rhPTH(1–84) (Jollette et al., 2006) in F344 rats. Despite these observations, an increased incidence of osteosarcomas has not been linked to the clinical use of either approved PTH product (Andrews et al., 2012; Cipriani et al., 2012; Krohn et al., 2016), though the duration of use has been limited to 2 years.

Based on its bone-anabolic action through agonism of the PTH1R, the carcinogenic potential of abaloparatide was examined in a 2-year study in F344 rats that followed the study design guidelines of the Committee for Medicinal Products for Human Use (CHMP) and OECD Guideline 417, and was reviewed by relevant regulatory agencies. In addition to the 3 dose levels of daily abaloparatide tested, a group of rats dosed with hPTH(1–34) at the mid-dose tested in the Vahle et al., 2004 paper (Vahle et al., 2004) was also included as a positive control group.

2. Methods

2.1. Study design

Male and female F344 rats were obtained from Charles River, St Constant, Quebec at 4–5 weeks of age and were acclimated for 3 weeks prior to initiation of treatment. Rats were housed individually in stainless steel cages prior to randomization, and group housed thereafter (up to 3 sex and group matched rats per cage)

except during designated study procedures. Certified pelleted commercial laboratory diet, Rodent Chow SCR4 (PMI Feeds, Inc. St Louis, MO) and municipal tap water (purified by reverse osmosis and ultraviolet irradiation) were provided ad libitum except during designated study procedures. This study was conducted at Charles River Laboratories Preclinical Services, Montreal in accordance with American Association of Laboratory Animal Care and Canadian Association of Laboratory Animal Care guidelines, was approved by Charles River Laboratories' Institutional Animal Care and Use Committee, and was conducted in compliance with the Good Laboratory Practice Regulations of the Food and Drug Administration (21 CFR Part 58).

Rats were randomly assigned to 5 treatment groups per sex ($n = 60/\text{group}$), stratified to body weight to ensure similar means across groups. Groups were administered abaloparatide at daily subcutaneous doses of 0, 10, 25, and 50 µg/kg; a separate group received 30 µg/kg hPTH(1–34) as a positive control. The abaloparatide doses of 10, 25, and 50 µg/kg were estimated to result in 3, 11 and 22-fold greater exposure than the 80 µg dose used in humans based on area-under-the-curve (AUC) predictions; the 30 µg/kg hPTH(1–34) dose used was approximately 14- to 25-fold higher than the 20 µg dose of teriparatide (FDA, 2002). SC injections (1 mL/kg body weight) were administered approximately the same time each day for 7 days a week, with injection sites rotated daily between 7 areas in the dorsal region.

Rats were observed twice daily for mortality and signs of ill health, with a detailed physical examination and food consumption and body weight recorded performed every week. From Week 26 onwards, all rats were also examined weekly for the presence of palpable masses. The site, size, and appearance of these masses were recorded when first detected and, following this initial description, their progression was monitored. Whole body radiography was performed on scheduled and unscheduled necropsies to detect abnormalities in the skeleton and alterations in soft tissue radiodensity to direct trimming of additional tissues for histopathology. When the number of surviving animals in any treatment group reached 20, test article was discontinued.

At necropsy, a full panel of tissues were collected and examined by histopathology, including the distal femur, proximal tibia, sternum, and lumbar vertebrae (L5–L6). In addition, all abnormalities detected by macroscopic and radiologic evaluation were subject to further microscopic evaluation. Except for those tissues with identified abnormalities, the right femur (including subregions) and the lumbar spine (L1–L4) were subjected to dual-energy x-ray analysis (DXA) ex vivo using a Hologic Discovery A scanner to quantify the pharmacodynamic response after up to 2 years of treatment.

2.2. Toxicokinetic (TK) and anti-therapeutic antibody analysis

Blood samples were collected from separate groups of animals dosed with 0, 10, 25, or 50 µg/kg abaloparatide or 30 µg/kg hPTH(1–34) SC daily ($n = 12/\text{sex/group}$) to capture drug exposure at 7 timepoints during the first 3 h after injection after 1, 26, and 52 weeks of dosing. Blood was obtained by jugular venipuncture without anesthesia, and the resulting plasma samples were analyzed for concentration of abaloparatide by radioimmunoassay at PPD Inc (ICDIM275.1 V3.00, Richmond, VA). hPTH(1–34) concentration was measured using a validated ELISA to confirm appropriate dosing. Pharmacokinetic parameters were estimated for the abaloparatide groups using WinNonlin pharmacokinetic software (Pharsight Corp., Mountain View, California). A non-compartmental approach consistent with the subcutaneous route of administration was used for parameter

Table 2

Mortality^a.

Sex	Male						Female					
	Vehicle	ABL	ABL	ABL	PTH	(µg/kg)	Vehicle	ABL	ABL	ABL	ABL	PTH
Group												
(µg/kg)	0	10	25 ^b	50 ^b	30 ^b		0	10	25	50	30	
N	60	60	59 ^c	60	60		60	60	60	61 ^c	60	
Mortality (all causes)	40	41	45 ^a	49 ^a	44 ^a		45 ^c	33	47	44 ^d	34	
Mortality (OSA-related)	0	19	32	35	25		0	4	8	24	8	

Pairwise comparison p-values to Group 1 (two-sided Peto's test): a = 0.0021, b < 0.0001, c = 0.0150, d = 0.0054.

Veh = vehicle, ABL = abaloparatide, PTH = human parathyroid hormone 1–34, OSA = osteosarcoma.

^a Data represent the number of animals that were moribund or required necropsy to the scheduled termination.

^b Male groups treated with ABL at 25 µg/kg, abaloparatide at 50 µg/kg and hPTH(1–34) at 30 µg/kg were terminated at weeks 97, 88 and 99, respectively when number of survivors reached 15/group.

^c One originally identified male was found to be a female pseudohermaphrodite at necropsy; it was reassigned to its true sex.

Other local radiological changes, either characterized by bone loss, bone production or mixed reaction (mixture of bone production and bone loss), were common observations in given abaloparatide or hPTH(1–34). Localized radiographic bone loss was observed at a number of skeletal sites, including the lumbar spine, sternum and tibia, in both male and female rats treated with abaloparatide at all dose levels, or hPTH(1–34), but rarely in vehicle control rats. Localized radiographic mineralization in soft tissues, predominantly in the lungs and abdominal organs, was a common change in rats administered abaloparatide or hPTH(1–34). This dose-dependent effect was more predominant in males than in females, and the histopathology of these changes are further described below.

3.4. Neoplastic finding in bone

Abaloparatide-related proliferative findings in bones included osteosarcoma with lesser frequency benign osteoblastoma and focal osteoblast hyperplasia, as summarized in Table 3. This continuum of effects on the osteoblast lineage was dose-dependent and present in both males and females administered ≥ 10 µg/kg abaloparatide. Similar proliferative changes were observed in hPTH(1–34)-treated rats at 30 µg/kg, with incidences similar to those observed with abaloparatide at 25 µg/kg. In contrast, bone neoplasia was uncommon in vehicle control rats with a single occurrence of osteosarcoma in males (tibia) and females (femur).

Osteosarcoma arose from various skeletal locations in abaloparatide-treated rats; the tibia was the most frequently affected single bone. Involvement of the forelimbs compared to the pelvic limbs was less common. Osteosarcoma was also frequently seen in the axial skeleton, mostly in the thoracic and lumbar spine. The neoplasm frequently impinged upon the spinal cord when located in the vertebral column (seen histologically in Fig. 2A), which provided an explanation for the hindlimb paralysis noted in some animals treated with abaloparatide or hPTH(1–34). Bones of the extremities (tail, feet, and skull) were generally spared by the neoplastic process. Overall, males were more susceptible to osteosarcoma development than females when treated with abaloparatide or hPTH(1–34). These neoplasms were often correlated with localized radiographic changes characterized by bone loss, bone production or mixed reaction, and/or masses found in bones at necropsy (Fig. 1).

Osteosarcomas were commonly multicentric, originating from different bones, and metastases to soft tissues were frequent and sometimes widespread to the lung, liver, spleen, adrenal gland, and kidney. These metastases often correlated with radiographic mineralization noted in lung and abdominal organs and, at

necropsy, with masses and/or nodules in the lung, liver and spleen. The microscopic appearance of the osteosarcomas were highly variable as previously reported with hPTH(1–34) (Vahle et al., 2002). The predominant histological types were the osteoplastic, osteoblastic and fibroblastic subtypes with a small number of telangiectatic osteosarcoma (Fig. 2). The compound subtype, combining bone and cartilaginous tumoral matrix, was rarely seen (Fig. 2E). The marked variation in amount of tumor bone between subtypes explains the radiological variability between osteosarcoma, which ranged from an osteolytic (bone loss) response to bone production.

In rats, osteoblastoma represents a benign neoplastic entity that has specifically been associated with chronic administration of rhPTH(1–34) or rhPTH(1–84) (Jollette et al., 2006; Vahle et al., 2002, 2004). Such neoplasm is characterized by an expansive endosteal and/or cortical growth that does not breach the periosteal surface, minimal invasion at the tumor margins and minimal cellular atypia and it is often associated with localized radiographic bone loss. Osteoblastoma was seen in various bones of abaloparatide-treated males and females at ≥ 10 µg/kg/day and most commonly affected the tibia; this tumor was also found in several hPTH(1–34)-treated males and females.

3.5. Other neoplastic findings

There were no other meaningful differences in the incidence of other neoplasms among male and female groups treated either with abaloparatide or hPTH(1–34). These neoplasms were those typically encountered in aging F344 rats and were randomly distributed among groups with no dose-related pattern identified (Table 4). The most commonly observed spontaneous tumors included interstitial (Leydig) cell adenoma in the testis, adenoma of the pars distalis in the pituitary gland (both sexes), large granular lymphocytic (LGL) leukemia (both sexes), benign pheochromocytoma in the adrenal gland (primarily in males), fibroadenoma in the mammary gland (primarily in females), endometrial stromal polyp in the uterus and C-cell adenoma in the thyroid gland (both sexes). Although the overall incidence of LGL leukemia was lower in males treated with abaloparatide and hPTH(1–34), this was likely related to the increased mortality in those groups.

3.6. Non-neoplastic findings in bone

Focal osteoblast hyperplasia (FOH) has been described as part of the morphologic continuum associated with chronic administration of rhPTH(1–34) or rhPTH(1–84) to the rat (Jollette et al., 2006; Vahle et al., 2002). This non-neoplastic finding was observed in abaloparatide-treated males and females with an increased

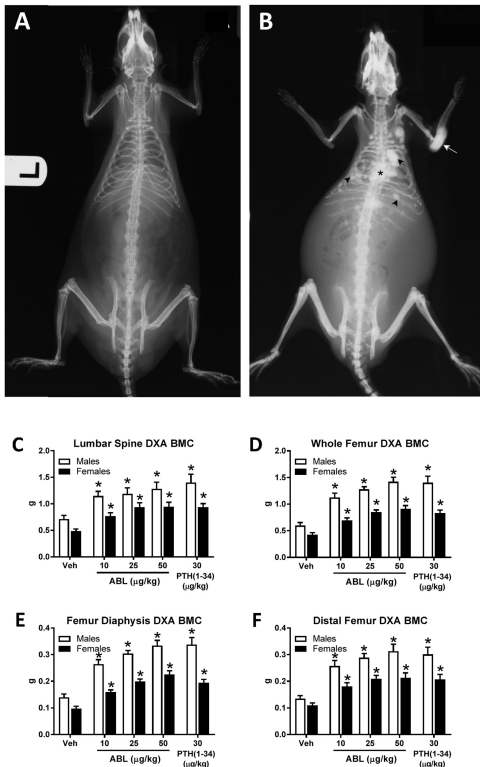


Fig. 1. Radiographic and densitometric changes in abaloparatide- and PTH-treated rats. Ventro-dorsal radiographs of males treated with (A) vehicle and (B) 25 µg/kg abaloparatide. Note in 1B the bilateral increased bone radiopacity in both axial and appendicular bones as well as several sites of localized bone production; radius (arrow), thoracic spine (*) and lung (arrowheads) confirmed histologically to be multicentric osteosarcoma with pulmonary metastasis. DXA BMC was measured *ex vivo* at the (C) lumbar spine (L1–L4) and (D) whole right femur, as well as in (E) diaphyseal and (F) distal regions of the femur. Mean $\pm$ SD, **p* < 0.05 vs Veh within each sex.

Table 4
Incidence of nonosseous neoplastic changes.

Sex	Male					Female				
Group	Veh	ABL	ABL	ABL	PTH	Veh	ABL	ABL	ABL	PTH
(µg/kg)	0	10	25	50	30	0	10	25	50	30
Animals examined	60	60	59	60	60	60	60	61	60	60
Testis/Interstitial cell adenoma	44	38	35	37	35	37	35	37	35	37
Pituitary gland/Adenoma: pars distalis	30	35	23	19	28	37	32	31	28	33
Systemic/LGL Leukemia	36	15	3	2	9	19	12	19	16	21
Adrenal gland/Phaeochromocytoma	10	8	7	3	4	4	2	1	28	36
Mammary gland/Fibroadenoma	2	—	—	—	—	17	22	20	15	11
Uterus/Endometrial stromal polyp	9	8	9	5	6	3	5	7	5	6
Thyroid gland/c-cell adenoma	9	8	9	5	6	3	5	7	5	6

Veh – vehicle, ABL – abaloparatide, PTH – human parathyroid hormone 1–34, LGL – Large granular lymphocytic.

incidence in males at ≥ 25 µg/kg/day, and in females at 10 µg/kg/day (see Fig. 3). FOH was also noted in some males and females administered hPTH (1–34). The tibia, lumbar vertebra and femur were the most frequent sites affected with FOH.

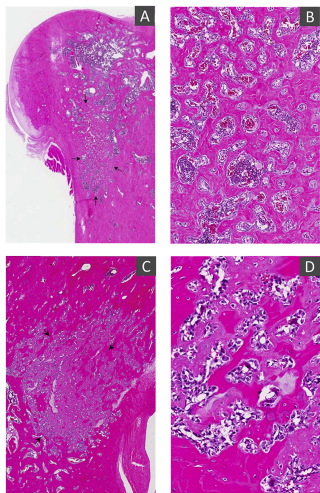


Fig. 3. Histologic features of non-malignant proliferative changes in abaloparatide-treated rats. (A) Well delineated osteoblast hyperplasia (arrows) with hyperostosis at periphery in the proximal humerus of a male treated with 50 µg/kg abaloparatide; (B) higher magnification in the central region illustrates the enlarged medullary spaces typically lined with an almost continuous rim of plump osteoblasts without excessive bone formation or lysis; (C) Osteoblastoma (arrows) with an expansive growth pattern and minimal invasion at the periphery in the distal femoral metaphysis of a male treated with 25 µg/kg abaloparatide; (D) higher magnification at tumor edge illustrates moderate number of large osteoblasts that are piling up with minimal cellular atypia. H&E.

Endosteal hyperostosis was a very common and systemic skeletal effect in abaloparatide-treated groups at ≥ 10 µg/kg/day (dose-dependent based on severity; data not shown) and in the hPTH (1–34)-treated groups. This change was characterized by trabecular hypertrophy, sometimes associated with peritrabecular fibrosis and/or woven bone deposition. The distribution was usually diffuse within the endosteal bone compartment and involved the epiphysis, metaphysis and diaphysis. Hyperostosis was seen at low incidence in the vehicle treated-rats, predominantly in the female subset; these observations were consistent with the reported spontaneous occurrence of this diagnostic entity in the aging F344 strain (Thurman and Bucci, 1994).

Osteofibrous dysplasia was another observed treatment-related effect in abaloparatide- and hPTH (1–34)-treated groups (Table 3). This microscopic change was scattered in various bones and sometimes correlated with focal radiographic bone loss. The incidence of this change was dose-dependent, with a similar number of observations across the 25 µg/kg abaloparatide and 30 µg/kg hPTH (1–34) groups in males (6) and females (10). Such lesion has been described in rhPTH (1–84)-treated rats (Jollette et al., 2006) and bears similarity with the focal stromal proliferation described by Vahle (Vahle et al., 2004) in bones of rats administered rhPTH (1–34). Osteofibrous dysplasia was characterized by replacement of the marrow and trabecular bone by well differentiated spindle cellular elements in a collagenous matrix admixed with predominantly woven small bone trabeculae lined with an osteoblastic rim. The fibroblastic and osteoblastic cells of this lesion lacked the cellular atypia present in their malignant counterparts.

3.7. Non-neoplastic/non-bone observations

Microscopic changes that occurred in both the abaloparatide and hPTH (1–34) treatment groups included effects on the large arteries, kidney, lung, spleen, and urinary bladder as reported in Table 5. These changes generally occurred at similar incidences across the 25 µg/kg abaloparatide and 30 µg/kg hPTH (1–34) groups. Medial mineralization, generally graded minimal to slight in severity with a multifocal distribution, involving the aortic arch and/or, less commonly, the pulmonary artery was commonly seen in animals treated with abaloparatide or hPTH (1–34). A dose response was not evident in either sex and males were more commonly affected than females. The change did not extend to other blood vessels including the routinely examined thoracic aorta. In the kidney, increased mineral localized in the pelvis was commonly observed in male and female rats given abaloparatide or hPTH (1–34). The incidence of this change was dose-dependent, with females were more commonly affected than males. The mineral deposits were generally graded minimal in severity and consisted in multifocal basophilic granular to finely spicular

Table 3
Neoplastic and non-neoplastic changes in bone.

Sex	Male ^a					Female ^a				
Group	Veh	ABL	ABL	ABL	PTH	Veh	ABL	ABL	ABL	PTH
(µg/kg)	0	10	25	50	30	0	10	25	50	30
N	60	60	59	60	60	60	61	60	60	60
Endosteal hyperostosis	4	60	59	60	60	11	59	61	60	60
Osteofibrous Dysplasia ^b	0	2	6	14	6	0	6	10	13	10
Osteoblast Hyperplasia ^b	1	5	15	21	9	1	11	4	6	4
Osteoblastoma	0	1	15a	20a	10b	0	8c	7d	9e	4
Osteosarcoma – primary	1	31a	46a	52a	39a	1	11f	22a	37a	24a
Osteosarcoma – metastasis	1	14	17	35	21	0	2	3	16	9
Osteosarcoma – multicentric	0	7	22	34	18	0	1	3	14	4

Pairwise comparison p-values to Group 1 (One-sided Peto's test): a < 0.0001, b = 0.0002, c = 0.0050, d = 0.0058, e = 0.0014, f = 0.0002. Veh – vehicle, ABL – abaloparatide, PTH – human parathyroid hormone 1–34.

^a The incidence listed refers to the number of animals affected with the finding in one or more bones.

^b Non-neoplastic findings were not subjected to inferential analysis.

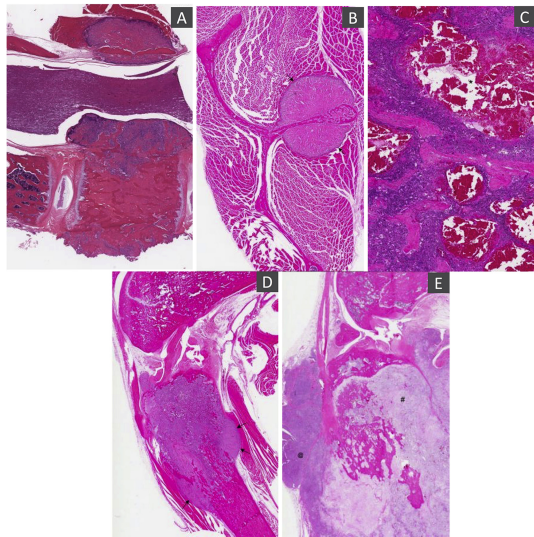


Fig. 2. Histologic features of osteosarcoma in rats administered abaloparatide or PTH for up to 2 years. (A) A mixed osteoblastic/osteoplastic osteosarcoma in a male administered 25 µg/kg abaloparatide; note the compression of the adjacent spinal cord. (B) The common osteoplastic variant from the scapula of a male treated with PTH. (C) The uncommon osteoblastic variant in the sacrum of a male treated with 10 µg/kg abaloparatide. (D) The common osteoblastic variant infiltrating the skeletal muscle (arrows) in the tibia of a male treated with PTH. (E) The uncommon compound variant with abundant cartilage (#) in the tibia of a male treated with 50 µg/kg abaloparatide. Neoplastic osteoblasts (Ø) have breached through the periosteum and invade the surrounding soft tissues. H&E.

Table 5
Other histopathologic observations in nonosseous tissues.

Sex	Male					Female					
Group	Veh	ABL	ABL	ABL	PTH	Veh	ABL	ABL	ABL	PTH	
(µg/kg/day)	0	10	25	50	30	0	10	25	50	30	
N	60	60	59	60	60	60	60	61	60	60	
Tissue	Observation										
Arteries ^a	Mineralization	4	29	33	35	37	0	6	12	11	5
Kidney: pelvis	Mineralization	1	14	27	28	28	1	23	37	41	33
Lung	Histiocytosis	16	30	28	27	31	23	37	40	38	36
Spleen	Hematopoiesis	5	14	29	39	22	8	29	33	31	30
Bladder	Dilatation	2	6	9	12	8	2	2	4	6	3
	Hemorrhage	1	4	7	9	6	0	1	4	3	0
	Inflammation	1	7	6	9	7	0	1	1	4	5

Veh – vehicle, ABL – abaloparatide, PTH – human parathyroid hormone 1–34.

Arteries affected were the aortic arch and, less often, pulmonary artery within the section of the heart.

material found free in the pelvic recess or trapped within or under the urothelium.

Increased extramedullary hematopoiesis in the spleen was noted at a greater incidence in animals treated with abaloparatide or hPTH (1–34) (Table 5), consistent with the marked reductions in marrow space observed histologically (as shown at the distal femur in Fig. 4) and secondary displacement of hematopoiesis. Dilatation, hemorrhage and/or inflammation of the urinary bladder were observed at a slightly increased incidence in abaloparatide and hPTH (1–34)-treated male rats. These observations were considered to be secondary, at least in some affected rats, to the presence of a malignant bone tumor localized in the spine or pelvis, which may have interfered with the neural control of micturition. Alveolar histiocytosis in the lung was slightly increased in incidence in males and females administered abaloparatide or hPTH (1–34). These changes were generally minimal in severity and multifocal and subpleural in distribution and may have occurred as a consequence to the frequent metastasis of osteosarcoma found in the lung.

4. Discussion

This study investigated the effects of near-lifetime exposure to the PTH1R agonist abaloparatide in F344 rats, comparing them to a 30 µg/kg dose of hPTH (1–34) which has previously been shown to induce osteosarcoma. Similar to hPTH (1–34), chronic daily abaloparatide administration resulted in a dose-dependent increase in the incidence of osteosarcoma and associated mortality. These effects occurred along with marked increases in bone mass, with

increased osteosarcoma evident at the lowest tested dose (10 µg/kg), representing an approximate 4-fold exposure multiple of the clinical dose. The incidence of neoplastic changes between abaloparatide and hPTH (1–34) at comparable exposure multiples (25 versus 30 µg/kg) were quite similar. Although these and other proliferative changes were observed in bone, abaloparatide did not increase the incidence of neoplastic changes in other tissues of the aged rat relative to vehicle or hPTH (1–34) treatment.

To date, an increased incidence of osteosarcoma has not been linked to the clinical use of either rhPTH (1–34) or rhPTH (1–84) (Andrews et al., 2012; Cipriani et al., 2012; Krohn et al., 2016), perhaps due to the differences in dosing (timing, duration, or level) or species (sensitivity to PTH1R agonists, modes of action in bone) between humans and rats. These chronic rat studies initiate PTH1R agonist administration during a phase of rapid longitudinal skeletal growth and are dosed for nearly their entire lifespan (Tashjian and Chahner, 2002). In contrast, PTH administration in postmenopausal women is initiated in an aged skeleton for only up to 2 years (2.5% of an 80 year lifespan). Rodents and primates, including humans, differ in their physiology and pharmacology of bone tissue. In addition to differences in intracortical remodeling (near-absent in rats), rodents have an exaggerated anabolic response to PTH and PTH1R and can tolerate much higher doses. The lowest doses at which rhPTH (1–34) (Vahle et al., 2002) and abaloparatide were tested in the 2-year rat model provided greater 3–4-fold greater exposure than the dose levels administered in humans. At these 'low' doses of hPTH (1–34) and abaloparatide, BMC was increased at the lumbar spine by approximately 50–60%, well above the observed changes over an 18-month period in postmenopausal

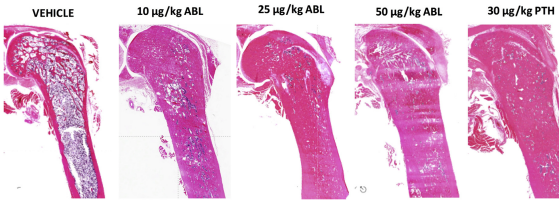


Fig. 4. Histologic increases in bone in the distal femur of abaloparatide-treated rats. Dose dependent increases in endosteal bone were evident in representative sagittal sections from the abaloparatide groups and 30 µg/kg PTH group relative to vehicle controls. H&E.

women treated with rhPTH(1–34) (Neer et al., 2001) or abaloparatide (Miller et al., 2016).

In addition to the current lack of observed clinical association between intermittent PTH administration and osteosarcoma to date despite ongoing pharmacovigilance (Andrews et al., 2012; Cipriani et al., 2012; Krohn et al., 2016), there are several clinical conditions that exhibit elevated PTH or PTHrP that have not shown an association with osteosarcoma. Patients with primary and secondary hyperparathyroidism did not have an increased incidence of osteosarcoma in the Swedish Cancer Registry database, nor did patients with activating mutations of PTH1R (eg. Jansen's metaphyseal chondrodysplasia) (Tschjahn and Chabner, 2002). To date, extended administration of rhPTH(1–34) or rhPTH(1–84) has not resulted in any reported osteosarcomas in patients with hypoparathyroidism, including children (Rubin et al., 2016; Theman et al., 2009; Winer et al., 2010). High levels of PTHrP have not been associated with increased incidence of osteosarcomas in patients with hypercalcemia of malignancy (Jibrin et al., 2006).

One theoretical concern regarding abaloparatide was based on its similarity to PTHrP, which was originally purified from multiple cancer cell lines and has been hypothesized to act in a paracrine fashion to mediate local resorption and facilitate tumor growth and metastasis (McCauley and Martin, 2012). However, some studies using human tissue have suggested that PTHrP positive tumors may exhibit reduced metastatic potential (Henderson et al., 2001, 2006), an effect distinct from its bone resorbing action. Conflicting evidence has been reported in mouse tumor models, with PTHrP expression both positively (Li et al., 2011) and negatively (Fleming et al., 2009) associated with initiation and progression of primary tumors. Although abaloparatide is distinct from both human PTHrP and hPTH(1–34), the data across published carcinogenicity studies indicate that chronic PTH1R agonism does not result in an increased incidence in tumors other than those originating from the osteoblast lineage.

In this study, a higher incidence of osteosarcoma and osteoblastoma was noted in the hPTH(1–34) treated groups when compared to the previous rat carcinogenicity studies evaluating the same 30 µg/kg dose of rhPTH(1–34) (Vahle et al., 2002). The use of radiographic evaluation of the entire skeleton to identify small tumors conducted in this study, (but not the prior studies with rhPTH(1–34)) likely increased the ability to detect small tumors and may explain the overall increased incidence in this study. Consistent with previous PTH rat carcinogenicity studies, osteosarcoma occurred at a higher rate in males than in females at similar dose levels of abaloparatide and hPTH(1–34). Drug concentrations were similar across sexes, and the cause of the increased osteosarcoma sensitivity in male rats remains unknown.

The non-neoplastic observations in this study including effects on the large arteries, kidney, lung, spleen, and urinary bladder have generally been reported in previous carcinogenicity studies with rhPTH(1–34) (FDA, 2012) or rhPTH(1–84) (Jolette et al., 2016). Although observations such as extramedullary hematopoiesis were clearly secondary to the changes in bone, the etiology of the other changes including soft tissue mineralization were less clear. These changes represent a class effect which was dose-dependent, with similar incidences, severity and tissue distribution between the abaloparatide and hPTH(1–34) groups at similar exposures.

5. Conclusion

The current study demonstrated an increased incidence of osteosarcoma in rats treated chronically with abaloparatide, consistent with previous studies with rhPTH(1–34). These changes were consistent across comparable doses of abaloparatide and hPTH(1–34), with histopathology reflecting similarities in both

neoplastic and non-neoplastic observations. Although the clinical translation of the increased rate of osteosarcoma with PTH1R agonists in rats remains unclear, these results suggest that there are no additional neoplastic risks associated with near-lifetime exposure to abaloparatide.

Conflict of interest statement

JJ, BA, AV, NM, ST, SYS are current or former employees and shareholders of Charles River Laboratories. GGL is an employee and shareholder of Experimental Pathology Laboratories, Inc. MSO and GH are employees and shareholders of Radius Health, Inc. MSO is a former employee and shareholder of Amgen, Inc.

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Transparency document

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TABLE 1.—Study design.			
Group number and description	Dose level (µg/kg/day)	Male	Female
		n	n
1. Vehicle control	0	60	60
2. Vehicle control	0	60	60
3. PTH – Low	10	60	60
4. PTH – Mid	50	60	60
5. PTH – High	150	60	60
6. PTH – High (DS) ^a	150	60	60

^aGroup 6 started dosing on study Week 24 (delayed start [DS]), and continued until study completion on Week 104. PTH = recombinant human parathyroid hormone 1–84.

osmosis and exposed to ultraviolet light, was freely available. The work was conducted in accordance with American Association of Laboratory Animal Care and Canadian Association of Laboratory Animal Care guidelines and was approved by the research laboratory Institutional Animal Care and Use Committee. Additionally, this study was conducted in compliance with the Good Laboratory Practice Regulations of the Food and Drug Administration (21 CFR Part 58).

Test Article and Study Design

Recombinant human PTH (NPS Pharmaceuticals Inc, Salt Lake City, UT) was administered as a solution in a citrate buffer containing mannitol. PTH doses of 10, 50 or 150 µg/kg body weight were administered daily by subcutaneous injection into the dorsal region for the 2-year period (Table 1). The high-dose (150 µg/kg/day) was expected to be the maximum tolerated dose based on a 6-month rat toxicology study and the low dose (10 µg/kg/day) was expected to provide a systemic exposure to PTH that was at least 2-fold greater than the systemic exposure in humans following a 100 µg dose. The study contained 2 groups of control rats that only received vehicle injections. A 6th group of rats (delayed start high-dose group; DS) was included to assess the skeletal response in the absence of PTH treatment during the period of active bone growth. This group began dosing (150 µg/kg/day) on Week 24 of the study and continued for the next 80 weeks. Dose volume was adjusted weekly for the first 16 weeks and monthly thereafter to account for changes in body weight.

All rats were observed twice daily for mortality and signs of ill health or reaction to treatment. In addition, a detailed physical examination was performed weekly for all rats. From Week 26 onwards, all rats were also examined weekly for the presence of palpable masses. The site, size, and appearance of these masses were recorded when first detected and, following this initial description, their progression was monitored.

Toxicokinetic Analysis

Blood samples were collected from satellite animals in Groups 2, 3, 4, and 5 at 8 time points on 4 separate days throughout the first 12 months of treatment. Blood was obtained by jugular venipuncture without anesthesia into tubes containing ethylenediaminetetraacetic acid. Plasma was stored at –80 °C until PTH analysis (Scantibodies Whole PTH Immunoradiometric Assay, Scantibodies Laboratory, Santee, CA). The animals in the satellite groups were euthanized after the 12-month blood sample was obtained.

Noncompartmental toxicokinetic analysis was performed on plasma PTH concentration data. Toxicokinetic analysis included assessment of the t_{max} , C_{max} , and area under

the curve, i.e., total systemic exposure. Area under the curve was calculated by the trapezoidal method (Gibaldi and Perrier, 1982) using WinNonlin, v3.2 (Pharsight Corporation, Mountain View, CA).

Radiographs

To enhance the detection of hidden or “occult” tumors that might otherwise have gone undetected at necropsy, radiographs of the whole skeleton (dorso/ventral and lateral planes) were performed under isoflurane anesthesia within 2 weeks of study end for all surviving carcinogenicity study rats. A Clinix-R (Picker International, Cleveland, OH) radiography machine was used with a Trimax 6 screen (3M, St. Paul, MN) and TMS/RA (Kodak, Rochester, NY) radiographic film. Radiographs were not taken for unscheduled necropsies. Each radiograph was examined by the study pathologist or veterinarian prior to necropsy in order to determine whether additional bones needed to be collected besides those listed in the protocol.

Necropsy

Animals euthanized on completion of treatment, as well as those euthanized for humane reasons, were anesthetized with isoflurane and exsanguinated from the abdominal aorta. All rats, including those found dead during the study, were necropsied and tissue samples were preserved.

A comprehensive collection of soft tissues was made, together with several routinely sampled bones: femur (distal left), tibia (proximal left), lumbar vertebrae (L5 and L6), sternum, and calvarium. Additionally, representative samples from any tissue with an external or internal macroscopic abnormality and bones with neoplasm-suspect radiological changes were collected. Collected tissues were fixed and preserved in 10% neutral-buffered formalin with the exception of ocular structures and male gonads, which were fixed in Zenker's fluid.

Bone Densitometry

The right femur and first four lumbar vertebrae (L1–L4) were also collected at necropsy, preserved overnight in 10% neutral-buffered formalin and transferred into 70% alcohol until densitometric scanning. These bones were collected from all main study animals euthanized at the end of treatment, with the exception of those bones with tumor-suspect macroscopic or radiological abnormalities, which were preferentially processed for histological evaluation. Bone mineral content (BMC) and bone mineral area (BMA) were measured by dual energy X-ray absorptiometry with a QDR-2000 plus bone densitometer (Hologic, Bedford, MA) and were used to derive bone mineral density (BMD). Measurements were made on segmented regions of the central and distal femur and the L1–L4 segments.

Histopathology

All specified tissues from all animals were prepared for histopathological examination by embedding in paraffin wax, sectioning and staining with hematoxylin and eosin. Bones were decalcified in dilute formic acid prior to processing. Preparation and evaluation of soft and bony tissues were done as follows:

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Defining a Noncarcinogenic Dose of Recombinant Human Parathyroid Hormone 1-84 in a 2-Year Study in Fischer 344 Rats

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ABSTRACT

The carcinogenic potential of human parathyroid hormone 1–84 (PTH) was assessed by daily subcutaneous injection (0, 10, 50, 150 µg/kg/day) for 2 years in Fischer 344 rats. Histopathological analyses were conducted on the standard set of soft tissues, tissues with macroscopic abnormalities, selected bones, and bones with abnormalities identified radiographically. All PTH doses caused widespread osteosclerosis and significant, dose-dependent increases in femoral and vertebral bone mineral content and density. In the mid- and high-dose groups, proliferative changes in bone increased with dose. Osteosarcoma was the most common change, followed by focal osteoblast hyperplasia, osteoblastoma, osteoma and skeletal fibrosarcoma. The incidence of bone neoplasms was comparable in control and low-dose groups providing a noncarcinogenic dose for PTH of 10 µg/kg/day at a systemic exposure to PTH that is 4.6-fold higher than for a 100 µg dose in humans. The ability of PTH to interact with and balance the effects of both the PTH-1 receptor and the putative C-terminal PTH receptor, may lead to the lower carcinogenic potential observed with PTH than reported previously for teriparatide.

Keywords: Parathyroid hormone; osteoporosis treatment; F344 rat; carcinogenicity; osteosarcoma; neoplasm; bone.

INTRODUCTION

Parathyroid hormone 1–84 (PTH) is the principal physiological regulator of systemic calcium and phosphate homeostasis. When injected on a daily basis, PTH and related analogs act through the PTH-1 receptor in bone cells to increase bone formation, mass and strength in animals and humans (Dempster et al., 1993; Fox, 2003). Teriparatide, the N-terminal 1–34 amino acid fragment of PTH, has been shown to prevent additional fractures in postmenopausal women with multiple fractures (Neer et al., 2001) and has been approved for the treatment of postmenopausal women with osteoporosis who are at high risk for fractures. However, the administration of teriparatide to Fischer 344 rats in carcinogenicity studies resulted in time- and dose-dependent increases in the incidence of proliferative lesions in bone, which included focal osteoblast hyperplasia, osteoblastoma, osteoma, and osteosarcoma (Vahle et al., 2002, 2004).

This result prompted the United States Food and Drug Administration to issue a draft guidance stating that the carcinogenic potential of PTH and related peptides should be evaluated in nonclinical studies (U.S. Food and Drug Administration, 2000). Prior to the initiation of such studies, it was hypothesized that chronic administration of PTH would be less likely to induce proliferative changes in bone than

teriparatide. PTH, unlike teriparatide, contains both the N-terminal region, which acts at the PTH-1 receptor to stimulate bone growth, and the C-terminal region which binds to a distinct PTH receptor that responds only to the C-terminal region of PTH (Iomata et al., 1995; Murray et al., 2005). This suggests a regulatory role for the C-terminal region of PTH in bone formation and is supported by a number of in vitro and in vivo studies demonstrating the opposing effects of the N- and C-terminal regions of PTH on alkaline phosphatase activity and apoptosis in bone cells, bone resorption and turnover, and plasma calcium levels (Murray et al., 1989; Jilka et al., 1999; Siatkowski et al., 2000; Divieti et al., 2001, 2002; Nguyen-Yamamoto et al., 2001; Langub et al., 2003). This hypothesis was tested in a carcinogenicity study using Fischer 344 rats that received daily subcutaneous injections of recombinant human PTH for 2 years.

MATERIALS AND METHODS

Animals and Husbandry

Male and female Fischer 344 rats (F344/NHsd) were obtained from Harlan Sprague-Dawley, Indianapolis, IN. At the initiation of the study the animals were 9–11 weeks old. Based on body weight, the rats were randomly assigned to treatment groups (n = 60/sex/group) and housed individually in stainless steel cages. To ensure that rats of each dose group were exposed for similar periods of time to different areas of the room, the animal cage racks were rotated once per month according to a predetermined schedule. Certified pelleted commercial laboratory diet, Rodent Chow 5002 (PMI Feeds, Inc, St. Louis, MO), was provided *ad libitum*. Municipal tap water, which had been softened, purified by reverse

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Abbreviations: PTH, parathyroid hormone 1–84; BMC, bone mineral content; BMA, bone mineral area; BMD, bone mineral density; DS, delayed start high-dose group.

- All tissues from all animals in both control groups and both groups dosed with 150 $\mu\text{g/kg/day}$
- All tissues from all animals in the 10 and 50 $\mu\text{g/kg/day}$ groups which died or were euthanized prior to scheduled necropsy
- All gross and radiological abnormalities in all groups
- Prior to study start, bones were specified as a target organ and were examined in all groups
- The spleen was identified as a target organ in the 150 $\mu\text{g/kg/day}$ female group and, thus, was examined in 10 and 50 $\mu\text{g/kg/day}$ female groups

The terminology and diagnostic criteria employed to interpret the various lesions seen in this study are broadly consistent with those contained within the Standardized System of Nomenclature and Diagnostic Criteria Guides for Toxicologic Pathology published by the Society of Toxicologic Pathologists (Long et al., 1995). The diagnostic criteria used to classify proliferative changes in bones are based on those used in the teriparatide carcinogenicity study (Vahle et al., 2002). Comparisons were also made with the human literature for some unusual bone lesions (Schwamm and Millward, 1995; Adler, 2000).

A single pathologist evaluated all tissues. Subsequently, a pathology peer review was performed to confirm the microscopic observations. The reported results reflect the mutually agreed-upon diagnoses by the primary and peer review pathologists, both certified by the American College of Veterinary Pathologists.

Statistical Analyses

Analyses were carried out to compare data from the main study test animals with data from each of the control groups and the combined control groups. Mortality data were analyzed using a Peto's 1-tailed trend test for most pairwise comparisons between groups (Peto et al., 1980); the analysis was performed using SAS Release 8.1 (SAS Institute, Cary, NC). Peto's 2-sided test was used to compare mortality for both control groups and both groups treated with 150 $\mu\text{g/kg/day}$. The significance of a linear dose-related increase in tumor occurrence rates was evaluated using Peto's survival adjusted 1-sided trend test. The analyses were performed in 3 phases; in turn, each control group and the combined control groups were considered. The overall trend test was performed excluding the 150 $\mu\text{g/kg/day}$ DS group and using the arithmetic dose level scores (0, 10, 50, and 150). Peto's 1-sided trend test was also used to test if the tumor rate in each treated group used for the trend test was significantly higher than the one in the combined control group (pairwise comparisons). Additional pairwise comparisons between both 150 $\mu\text{g/kg/day}$ groups were performed using Peto's 2-sided trend test.

Each trend test (the overall trend and the pairwise comparison tests) of the tumor data was conducted at the 5% significance level and was interpreted according to the recommendations of Lin and Rahman (1998).

The 1-sided Cochran-Armitage trend test followed by Fisher's exact test for comparison between groups was used to analyze nonneoplastic lesion data. The dual energy X-ray absorptiometry data were analyzed with Levene's test and, if the outcome was significant ($p \leq 0.05$), data were subjected to a Kruskal-Wallis test followed by a Wilcoxon-rank-sum test.

TABLE 2.—Systemic exposure to PTH.*

Dose ($\mu\text{g/kg/day}$)	AUC (ng-hr/mL)			Rat:human exposure ratio
	Month 6	Month 12	Mean	
0	0.04	0.04	0.04	—
10	3.65	3.75	3.70	4.6
50	22.1	20.9	21.5	27
150	48.3	58.0	53.2	66

*Combined gender data.
*The AUC value for the control group was estimated using the lower limit of the assay (15.25 pg/mL) over 150 minutes postdose.

Exposure ratio = $\text{AUC}_{\text{rat}}/\text{AUC}_{\text{human}}$, where $\text{AUC}_{\text{human}} = 0.8$ ng-hr/mL, in post-menopausal osteoporotic women receiving 100 μg of PTH for 12–15 months. An assay with high specificity for full-length human PTH was used in both species.

AUC = area under the curve; PTH = recombinant human parathyroid hormone 1–84.

Analysis of variance and *t*-tests on least-square means were carried out if the outcome of Levene's test was not significant. All references to statistical significance refer to comparisons to the combined control groups, unless otherwise noted.

RESULTS

Systemic Plasma Drug Levels

Total systemic exposure to PTH at steady-state was reflected by the 6- and 12-month area under the curve measurements, the mean of which was used for comparison with human exposure (Table 2). The low dose provided a systemic exposure to PTH that was 4.6-fold greater than the systemic exposure in humans following a 100 μg dose. Although not evaluated in this study, no anti-PTH antibody formation was observed in a separate 6-month rat toxicology study employing PTH doses similar to and greater than the doses used in this study (unpublished observations). This finding and the absence of a decline in systemic exposure over the 12 months of toxicokinetic sampling in the current study, suggest that there is no immune response by the rat to human PTH.

Mortality

There was an increased rate of mortality at Week 94 in Group 5 males receiving 150 $\mu\text{g/kg/day}$ compared with the combined vehicle control groups (67% versus 30%, respectively) leading to discontinuation of treatment for Group 5 males at the end of Week 94. For females, there was no significant effect of PTH treatment on mortality (Table 3). The mortality rate for males was notably higher than in females, irrespective of treatment status and was attributed to the greater severity of chronic progressive nephropathy, which was the most common fatal, nonneoplastic condition in males. An

TABLE 3.—Survival in rats treated with PTH for 2 years.

Group Identification	Survival (%)	
	Males	Females
1. Vehicle control	33	58
2. Vehicle control	27	57
3. PTH 10 $\mu\text{g/kg/day}$	28	45
4. PTH 50 $\mu\text{g/kg/day}$	28	35
5. PTH 150 $\mu\text{g/kg/day}$	N/A	42
6. PTH 150 $\mu\text{g/kg/day}$	30	52

*Delayed start of dosing to study Week 24.
PTH = recombinant human parathyroid hormone 1–84; N/A = Not available due to early termination at Week 94.

TABLE 4.—Radiologic findings in surviving animals at study completion.

Group	Gender									
	Male					Female				
PTH dose ($\mu\text{g/kg/day}$)	1	2	3	4	5	6	1	2	3	4
Number of animals subjected to radiography	0	0	10	50	150	150*	0	0	10	50
X-ray findings, % of affected animals ^b										
Bone sclerosis	4	0	38	95	100	100	3	9	25	100
Bone loss	0	6	0	21	6	37	6	6	14	29
Bone production	0	0	0	5	31	11	0	7	4	8
Mixed reaction	0	0	0	16	25	5	3	0	0	16
Occult neoplasms, no. of tumors ^c	0	0	0	1	0	0	0	0	1	4 ^d

*DS = Delayed start of dosing on study Week 24.

^bUsing the number of animals radiographed as denominator.

^cHistologically confirmed primary bone neoplasms (mostly osteosarcomas) where radiography was essential for diagnosis.

^dIncluded 1 fibrosarcoma.

^eIncluded 1 osteoblastoma.

PTH = recombinant human parathyroid hormone 1–84.

incidence of approximately 100% in the $\geq 50 \mu\text{g/kg/day}$ dose groups (Table 4). In bones of the appendicular skeleton, the sclerosis was generally bilateral, affecting most commonly the femurs and tibiae. In the femurs, the typical presentation was a thickening of the cortices with a concurrent reduction of the medullary space. Both the spine and skull were often affected in the axial skeleton. Bone sclerosis was occasionally observed in the control groups. However, the skeletal distribution of sclerosis in control animals was generally not as widespread as in PTH-treated rats.

Localized and generally mono-ostotic radiological changes, either characterized by bone production, bone loss or a mixed reaction (both bone production and loss), were common observations in PTH-treated groups (Table 4). Overall, radiological changes were more frequently seen in the 50 or 150 $\mu\text{g/kg/day}$ groups than in the 10 $\mu\text{g/kg/day}$ and control groups. These changes were used to identify additional bones for histopathologic evaluation.

Radiological assessments made possible the identification of additional small bone tumors, approximately 10% overall and approximately 40% in the 150 $\mu\text{g/kg/day}$ female DS group, at sites not routinely sampled at necropsy and that would otherwise have gone undetected. These tumors were referred to as "occult" neoplasms (Table 4). Radiographs also proved a useful complement to support the histological diagnosis of some neoplasms.

Neoplastic Histological Observations

Several types of bone neoplasms were observed in both males and females treated with PTH at $\geq 50 \mu\text{g/kg/day}$. Most neoplasms were regarded as malignant and classified as osteosarcoma. Other skeletal neoplasms included malignant fibrosarcoma, benign osteoblastoma, and osteoma. The incidence of all bone neoplasms was significantly increased in the whole skeleton of both males and females treated at $\geq 50 \mu\text{g/kg/day}$ (Groups 4 and 5). In groups treated with vehicle or PTH at 10 $\mu\text{g/kg/day}$, primary bone tumors were rare and limited to the occurrence of osteosarcomas in 2 female control animals and 1 male in the 10 $\mu\text{g/kg/day}$ group (Table 5). No statistically significant differences in the incidence of bone neoplasms were found between males and females treated with PTH at 10 $\mu\text{g/kg/day}$ and their respective control groups, consistent with the 10 $\mu\text{g/kg/day}$ dose having no tumorigenic effect on bones.

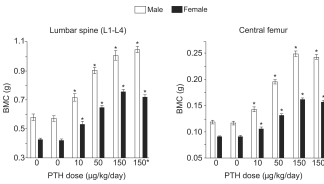


FIGURE 1.—Effects of daily treatment for 2 years to Fischer 344 rats on bone mineral content (BMC) at the lumbar spine and central femur. Values are mean $\pm$ standard error. * $p < 0.01$; significantly different from combined control groups. 150^c: Delayed start of dosing on Study Week 24. PTH = recombinant human PTH 1–84.

increased incidence of fatal osteosarcoma was noted in males treated at 50 $\mu\text{g/kg/day}$, and males and females treated at 150 $\mu\text{g/kg/day}$ in Groups 5 and 6. The first osteosarcoma fatalities were seen at Weeks 58, 55 and 65 in a control female, a 150 $\mu\text{g/kg/day}$ DS male and a 150 $\mu\text{g/kg/day}$ male, respectively.

Radiology

Bone sclerosis, defined by an increased diffuse radiopacity, was commonly observed in all PTH groups (Figure 2) with an

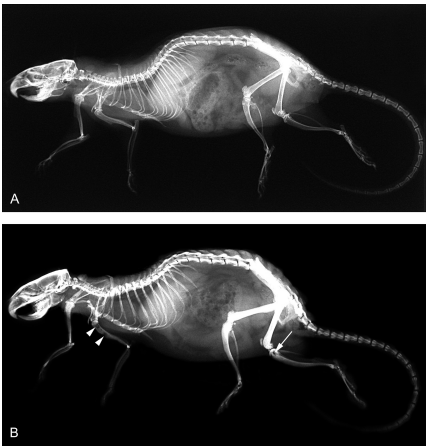


FIGURE 2.—Lateral view of whole body radiographs. (A) Control female. (B) DS female treated for 13.5 months with 150 $\mu\text{g/kg/day}$ PTH shows generalized increased radiopacity of the skeleton, which is most evident in femurs with obliteration of the medullary cavity, and localized bone loss in humerus (arrowheads) and tibia (arrow) subsequently diagnosed as multicentric osteosarcoma.

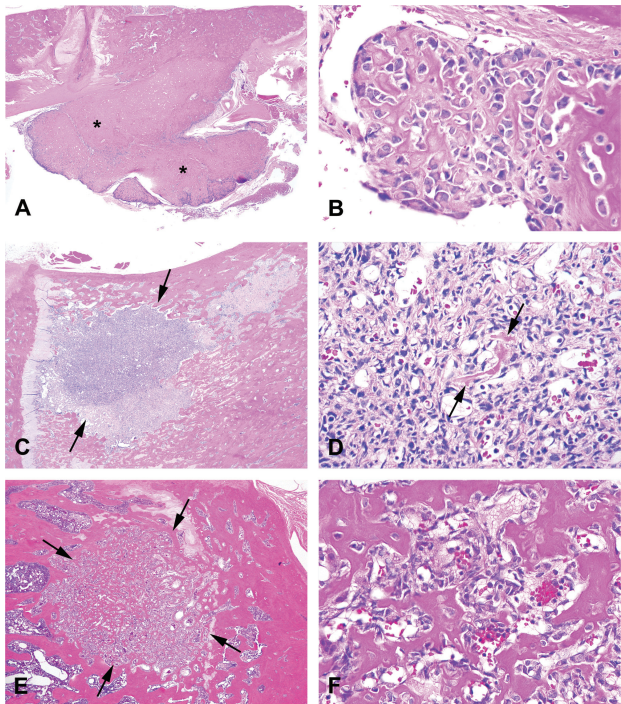


FIGURE 3.—Common histological features of PTH-induced neoplasms. (A, B) Osteoplastic osteosarcoma in thoracic vertebral body of a female treated for 24 months with 150 $\mu\text{g/kg/day}$ where the zonation phenomenon is evident with centrally located tumor bone (*) and anaplastic osteoblasts at outer margin (B). (C, D) Fibroblastic osteosarcoma (C, arrows) in tibia of a DS female treated for 13.5 months with 150 $\mu\text{g/kg/day}$. This lytic tumor was initially tracked with radiography and produced scant tumor osteoid (D, arrows). (E, F) Well-defined osteoblastoma (E, arrows) in proximal tibia of a male treated 22 months with 50 $\mu\text{g/kg/day}$ H&E.

There was a significant, dose-related trend in the occurrence of osteosarcomas. Osteosarcomas, combined for all sites, were significantly increased in males treated with PTH at 50 $\mu\text{g/kg/day}$ and males and females treated with 150 $\mu\text{g/kg/day}$. At individual bone sites, a significant increase in

osteosarcomas was observed in the lumbar vertebrae, tibia, and femur of males in the 150 $\mu\text{g/kg/day}$ group and in the tibia of males treated with 50 $\mu\text{g/kg/day}$. High-dose PTH was also associated with a significant increase of osteosarcomas in lumbar vertebrae of female rats. Delaying the start of

TABLE 5.—Noteworthy tumors found in all examined bones.*												
Group (n = 60/group) Dose (μg/kg/day)	Sex											
	Male						Female					
	1	2	3	4	5	6	1	2	3	4	5	6
0	0	0	10	50	150	150 ^b	0	0	10	50	150	150 ^c
Number of animals affected with one or more:												
Osteoblastoma	0	0	0	2	4 ^e	3	0	0	0	3 ^e	9 ^e	1
Osteoma	0	0	0	1	2 ^e	0	0	0	0	1	1	0
Osteosarcoma	0	0	1	13 ^e	27 ^e	13 ^d	2	0	0	5	13 ^e	20
Fibrosarcoma	0	0	0	1	0	0	0	0	0	1	0	2
All bone neoplasms	0	0	1	17 ^e	30 ^e	14 ^d	2	0	0	10 ^e	21 ^e	21
Number of tumors												
Primary tumors ^f	0	0	1	17	45	17	2	0	0	11	25	26
Multicentric neoplasms ^g	0	0	0	0	0	3	0	0	0	1	4	5
Metastatic osteosarcoma	0	0	1	6	14	5	1	0	0	1	7	7

*Includes metastatic osteosarcomas found only in soft tissues that were considered to originate from an undetermined skeletal origin.
^bDelayed start of dosing on study Week 24.
^cDifferent from combined control groups ($p < 0.05$; Peto's test).
^dDifferent from Group 5 ($p < 0.05$; Peto's test).
^eCombination of above listed tumors.

PTH treatment for 6 months resulted in a significantly lower bone tumor incidence in males, but not in females (Group 6 versus 5).

Fibrosarcoma

This rare, malignant mesenchymal neoplasm of the skeleton was only seen in 4 animals treated with PTH at ≥ 50 $\mu\text{g/kg/day}$ (Table 5). The distal femur was the primary site affected with a single involvement of the lumbar vertebrae. These fibrosarcomas were small, confined within the bone, and were detected during radiological examination as sharply localized focal osteolysis. A dense proliferation of anaplastic, usually fusiform, mesenchymal cells that lacked the production of any tumor osteoid matrix and that replaced preexisting bone, were the criteria used for histological diagnosis (Figure 5). None of the statistical analyses conducted with this tumor reached significance.

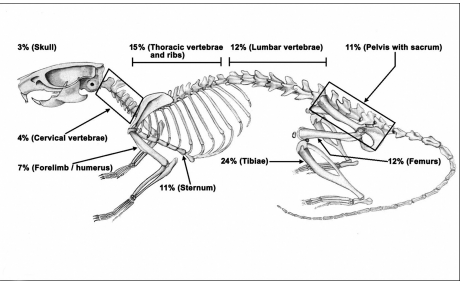


FIGURE 4.—Gender-combined primary skeletal localization of osteosarcoma in rats treated with PTH (based on 105 tumors).

TABLE 6.—Number of animals with relevant nonneoplastic histological findings found in bones.												
Group (n = 60/group) Dose (μg/kg/day)	Sex											
	Male						Female					
	1	2	3	4	5	6	1	2	3	4	5	6
0	0	0	10	50	150	150 ^a	0	0	10	50	150	150 ^a
Osteosclerosis												
Femur	1	0	26	58	60	60	14	12	44	60	60	60
Lumbar vertebra	2	2	14	51	60	60	11	10	43	59	60	60
Sternum	0	2	10	57	60	60	18	13	38	59	60	60
Tibia	0	0	15	58	60	60	11	11	41	60	60	59
Fibrous osteodystrophy												
Femur	1	0	0	1	1	1	1	1	2	3	6	10
Lumbar vertebra	2	0	0	6	13	15	0	1	1	9	17	17
Sternum	0	0	0	4	2	5	1	0	1	3	14	12
Tibia	1	0	0	3	2	0	0	0	1	3	3	8
Focal osteoblast hyperplasia												
Femur	0	0	0	0	3	1	0	1	0	0	6	2
Lumbar vertebra	0	0	0	0	2	1	0	0	1	4	3	5
Sternum	0	0	0	0	1	2	0	0	0	0	2	2
Tibia	1	2	0	4	8	1	0	2	0	0	2	4
All bone sites	1	2	0	5	12	5	1	3	1	4	12	13
Osteofibrous dysplasia												
Femur	0	0	0	0	0	0	0	0	0	1	3	3
Lumbar vertebra	0	0	0	0	0	0	0	0	0	0	1	1
Sternum	0	0	0	0	0	0	0	0	0	1	0	1
Tibia	0	0	0	0	0	0	0	0	1	1	1	1

^aDS = Delayed start of dosing to study Week 24.

focal osteoblast hyperplasia at all sites in males and females treated with 150 $\mu\text{g/kg/day}$ and at the lumbar vertebrae in females treated with 50 $\mu\text{g/kg/day}$ (Table 6). In male rats, the most commonly affected bone was the tibia, while the femur and lumbar vertebrae were most commonly affected in females. Focal osteoblast hyperplasia was a rare observation in animals dosed with 10 $\mu\text{g/kg/day}$ and, when all bones were evaluated together, its incidence in this dose group was comparable to or slightly less than in control animals.

Osteofibrous Dysplasia. Osteofibrous dysplasia, localized primarily in the femur, was another microscopic change seen in several female rats dosed at ≥ 50 $\mu\text{g/kg/day}$, but not in the control or 10 $\mu\text{g/kg/day}$ groups or in any males (Table 6). The histology of these lesions resembled its human counterpart (Schwamm and Millward, 1995) and was characterized by replacement of marrow and trabecular bone by well-differentiated spindled cellular elements in a collagenous matrix admixed with predominantly small woven bone

Osteoblastoma

Osteoblastoma is a benign osseous neoplasm which was first reported in rats treated with teriparatide (Vahle et al., 2002). This tumor typically contains a moderate number of large and active osteoblasts along bone trabeculae. Diagnostic hallmarks used to identify an osteoblastoma included an intramedullary expansive growth pattern with minimal invasion at the margins, minimal cytological atypia, and scattered normal-looking mitotic figures (Figure 3). Only animals treated with PTH at ≥ 50 $\mu\text{g/kg/day}$ were diagnosed with osteoblastoma (Table 5), which were mainly found in the tibia, femur, lumbar vertebra and sternum. When analyzed across all examined osseous sites, significant increases in incidence were noted for females treated at 50 $\mu\text{g/kg/day}$ and males and females at 150 $\mu\text{g/kg/day}$ (Group 5). In several animals, some tumors in the same section consisted of a portion typical of an osteoblastoma that was adjacent to cells with a malignant phenotype. These dual occurrences were

trabeculae lined with an osteoblastic rim. In contrast to fibrous osteodystrophy, these lesions presented with substantial bone resorption, which was often seen on radiographs as focal bone loss. The fibroblastic and osteoblastic cells of fibrous osteodystrophy and osteofibrous dysplasia lacked anaplastic features, which were used to differentiate them from bone malignancies.

Miscellaneous Nonneoplastic Histological Observations

Extraskelatal effects associated with PTH treatment were noted in a few soft tissues. The incidence of increased extramedullary hematopoiesis was higher in females treated at ≥ 50 $\mu\text{g/kg/day}$. This dose-related change was considered secondary to the marked osteosclerosis that displaced hematopoietic production from the obliterated bone marrow to the spleen. Dilatation and/or inflammation of the urinary bladder was more common in both male groups treated with 150 $\mu\text{g/kg/day}$. These findings were regarded as secondary,

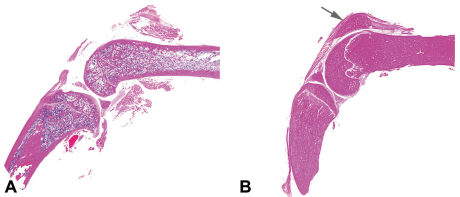


FIGURE 6.—Sub-gross sections of femoro-tibial joint. (A) Control female. (B) Severe diffuse osteosclerosis with marrow obliteration noted in femur, tibia and patella (arrow) of a female treated for 24 months with 150 $\mu\text{g/kg/day}$ PTH. H&E.

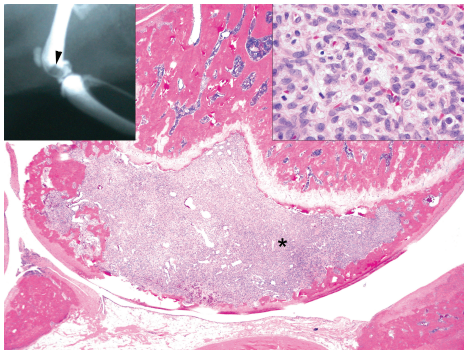


FIGURE 5.—Intra-medullary fibrosarcoma in distal femur (*) with metaphyseal osteosclerosis in a female rat treated 24 months with 50 $\mu\text{g/kg/day}$ PTH. Note radiographic focal osteolysis of femoral epiphysis (arrowhead) with bone sclerosis of metaphysis and diaphysis (left insert). The anaplastic cells were not producing tumor osteoid (right insert). H&E.

diagnosed solely as osteosarcomas, suggesting neoplastic progression.

Osteoma

This very rare benign tumor in rats was only seen in animals treated with PTH at ≥ 50 $\mu\text{g/kg/day}$, where it was localized to the skull, lumbar vertebrae and ilium. When the osteomas were analyzed across all examined skeletal sites, a significant increase was found for males treated at 150 $\mu\text{g/kg/day}$ (Group 5; Table 5).

Miscellaneous Neoplastic Findings

PTH treatment did not increase the incidence of other neoplasms in any of the soft or skeletal tissues examined.

Nonneoplastic Histologic Observations

Osteosclerosis. Osteosclerosis was commonly seen in males and females given 10 $\mu\text{g/kg/day}$ PTH and it affected almost all routinely examined bones from animals given ≥ 50 $\mu\text{g/kg/day}$ (Table 6) confirming the widespread distribution seen with radiography. This change was primarily characterized by trabecular hypertrophy, which increased the volume of bone occupying the medullary cavity (Figure 6). The distribution was usually diffuse within the endosteal compartment and involved the epiphyseal, metaphyseal and diaphyseal regions in long bones. Often, bone accumulation in the medullary cavity was sufficiently severe to almost entirely obliterate the space occupied by the bone marrow. Osteosclerosis was regarded as an exaggerated pharmacological response to treatment with PTH.

Osteosclerosis was also observed in control rats, especially in females (Table 6). The low incidence of osteosclerosis was

consistent with spontaneous hyperostosis or osteopetrosis; a condition reported predominantly in aging female Fischer 344 rats (Leininger and Riley, 1990; Thurman and Bucci, 1994). This observation also correlated with the radiological bone sclerosis noted in a small number of control rats. Nonetheless, the incidence of osteosclerosis was significantly lower in both male and female control groups compared with the 10 $\mu\text{g/kg/day}$ dose group at each of the routinely sampled bone sites.

Fibrous Osteodystrophy. Fibrous osteodystrophy, predominantly noted in the lumbar vertebrae and sternum, was a finding seen predominantly in males and females treated at ≥ 50 $\mu\text{g/kg/day}$ (Table 6). These microscopic, focal or multifocal lesions in the spongiosa were generally admixed with osteosclerosis and consisted of increased numbers of bone cells lining a woven bone matrix and of stromal cells and fibrous connective tissue in the adjacent marrow cavity (Figure 7). Such lesions were differentiated from those resulting from renal osteodystrophy, a complication of the advanced chronic progressive nephropathy seen in many control and PTH-treated males. Prominent osteoclastic resorption, a feature of renal osteodystrophy, was lacking in animals diagnosed with fibrous osteodystrophy. Overall, the sporadic occurrence of fibrous osteodystrophy in males and females was comparable between the control and 10 $\mu\text{g/kg/day}$ PTH groups.

Focal Osteoblast Hyperplasia: Focal osteoblast hyperplasia is characterized by microscopic focal proliferation of well-differentiated osteoblasts filling enlarged medullary spaces without excessive bone lysis or formation (Figure 7). PTH treatment was associated with an increased incidence of

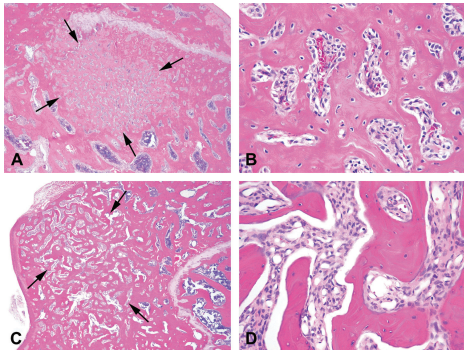


FIGURE 7.—Non-neoplastic PTH-induced changes. (A, B) Focal osteoblast hyperplasia blends in the osteosclerotic femoral metaphysis of a male treated for 21 months with 150 $\mu\text{g/kg/day}$. The remodeled delicate trabecular architecture allows spotting hyperplasia at low power (arrows) and is lined with columnar but well differentiated osteoblasts. (C, D) Fibrous osteodystrophy in the osteosclerotic femoral epiphysis of a female treated for 20 months with 50 $\mu\text{g/kg/day}$. A well-differentiated fibrovascular stroma (arrows) with paucity of active osteoblasts and osteoclasts focally replaces the hematopoietic marrow. H&E.

at least in some treated rats, to a malignant bone tumor localized in the vertebrae or pelvis.

DISCUSSION

This study characterized the effects of chronic, near lifetime exposure to recombinant human PTH on the rat skeleton in a cancer bioassay. A PTH dose of 10 $\mu\text{g/kg/day}$ for 2 years significantly increased bone mass, but did not increase the incidence of osteosarcoma or induce other neoplastic and non-neoplastic proliferative changes. This dose provided a systemic exposure to PTH that was 4.6-fold greater than the systemic exposure in humans following a 100 μg dose. A dose-related increase in osteosarcoma was observed with the 50 and 150 $\mu\text{g/kg/day}$ doses of PTH.

Fibrosarcoma, another malignant skeletal tumor, was observed in four of 360 rats exposed to ≥ 50 $\mu\text{g/kg/day}$. While the teriparatide oncogenicity studies did not report such tumors (Vahle et al., 2002, 2004), the introduction of radiography in this study allowed detection and sampling of these small lesions. This tumor was assumed to be associated with PTH treatment.

The incidence, skeletal distribution and histological appearance of osteoblastoma and osteoma resulting from mid- and high-dose PTH administration were similar to teriparatide (Vahle et al., 2002, 2004). However, these benign bone tumors were not observed in the low-dose PTH group.

Focal osteoblast hyperplasia has been described as part of the morphologic continuum of hyperplasia leading to neoplasia associated with near-lifetime administration of teriparatide to the rat (Vahle et al., 2002). It was observed in this

study with comparable incidence in control and low-dose PTH animals and with higher incidence in animals dosed with PTH at the mid- and high-dose.

Fibrous osteodystrophy and osteofibrous dysplasia were observed in this study, but were not previously observed following chronic teriparatide exposure (Vahle et al., 2004). However, focal stromal and stromal-vascular proliferations reported following long-term exposure to teriparatide (Vahle et al., 2004) were not observed in this study. The incidence and histological appearance of focal stromal and stromal-vascular proliferations following teriparatide administration and of fibrous osteodystrophy and osteofibrous dysplasia in this study appeared to be similar, suggesting that they represent the same entities. Our selection of a different diagnostic terminology reflects our opinion that these lesions are more than just stromal proliferation because they include some bone loss and abnormal woven matrix deposition.

The DS group treated with 150 $\mu\text{g/kg/day}$ PTH from 8 months to 2 years of age, was included to assess the skeletal response in absence of PTH treatment during the period of active bone growth. In both genders the delayed start of high-dose PTH was also associated with a high incidence of osteosarcomas suggesting that rapid longitudinal bone growth is not necessary for the induction of neoplasms at this high dose.

While this study did not directly compare PTH with teriparatide, it was designed to allow comparisons between the effects of both peptides. As in the teriparatide study (Vahle et al., 2002), this bioassay used the same inbred strain of Fischer 344 rats that were fed ad libitum, kept in a similarly controlled environment and exposed for the same time

period and by the same route of administration at similar molar dose levels. Furthermore, the diagnostic criteria used for the classification of most proliferative lesions associated with teriparatide were applied to the PTH study. However, unlike the teriparatide study, radiographs were used to detect small tumors at skeletal sites not routinely sampled at necropsy and that otherwise would have gone undetected. This increased the number of skeletal neoplasms identified by 10% and supports previous recommendations that radiography be used to detect bone tumors in oncogenicity studies (Stanton, 1979; Zwicker and Eyster, 1996).

In this animal model, the responses of the skeleton to PTH and teriparatide were similar in many ways (Vahle et al., 2004). Like teriparatide, PTH induced a significant dose-dependent increase in bone mass with all doses at all skeletal sites evaluated. This pharmacological effect resulted in a generalized osteosclerosis observed by radiology and histology. At the higher doses, these changes were often so marked that the marrow was almost completely obliterated. Also like teriparatide, the administration of PTH for 2 years resulted in dose-related proliferative changes in bone where osteosarcoma predominated. However, in contrast to teriparatide, the lowest dose of PTH was not carcinogenic with the incidence of osteosarcoma equal to that observed in control animals. The histological features of the osteosarcomas in the control and low-dose PTH groups were similar.

Collectively, the data support the concept that induction of skeletal neoplasia associated with PTH is dose-dependent, but that the threshold for bone tumor development occurs at a higher dose than that required to obtain the desired pharmacological effect. Our data also suggest that at exposures similar to those observed clinically, the oncogenic potential of PTH is lower than teriparatide in comparable 2-year rat carcinogenicity studies. When compared to 24 months of treatment with teriparatide (Vahle et al., 2002), treatment with PTH resulted in a notable right-shift in the dose response curve for osteosarcomas (Figure 8). This affords a safety margin for PTH of at least 4.6-fold between exposures following the clinically recommended dose for humans (100 μ g/day) and the demonstrated noncarcinogenic dose administered to rats (10 μ g/kg/day). In contrast, the lowest dose of teriparatide

evaluated in rats (5 μ g/kg/day) resulted in an exposure only 3 times that observed with the 20 μ g/day clinical dose and was associated with an increased incidence of osteosarcoma (Eli Lilly and Company, 2004).

The reason long-term exposure to PTH is associated with a lower incidence of osteosarcoma than teriparatide is unknown, but it is possible that the C-terminal region of PTH, which is absent in teriparatide, may be responsible. C-terminal fragments of PTH are secreted by the parathyroid glands and produced following peripheral PTH metabolism, primarily in the liver (Nguyen-Yamamoto et al., 2002). It is becoming increasingly recognized that a protein that binds PTH and C-terminal PTH fragments, but not teriparatide, is present in bone cells (Murray et al., 2005).

Activation of this putative C-terminal PTH receptor elicits biological responses that are often in opposition to those that occur following PTH-1 receptor activation. For example, C-terminal PTH fragments blunt the increase in serum calcium levels induced by teriparatide and PTH infusions in thyroparathyroidectomized rats (Slatopolsky et al., 2000; Nguyen-Yamamoto et al., 2001; Langub et al., 2003). Additionally, teriparatide stimulates bone resorption while C-terminal PTH fragments inhibit bone resorption. Finally, teriparatide inhibits apoptosis in bone cells while C-terminal PTH fragments stimulate apoptosis in bone cells (Bringham, 2002).

It is this latter phenomenon that provides a possible explanation for the difference in oncogenicity between the 2 peptides; the N-terminal and C-terminal regions of PTH may act in concert to control and maintain a normal rate of osteoblast turnover. PTH-1 receptor activation in the absence of such feedback regulation could lead to osteoblast hyperplasia and neoplastic transformation. A similar regulatory mechanism has been described in the prolactin/growth hormone endocrine system where the full-length hormones stimulate proliferation of capillary endothelial cells in vitro and in vivo, but N-terminal peptides derived from these hormones are inhibitory and act through a distinct receptor (Struman et al., 1999).

In conclusion, recombinant human PTH represents a potentially new treatment for individuals with osteoporosis who are at a risk of fracture. In the rat carcinogenicity study reported here, treatment with PTH resulted in significant new bone growth at all the evaluated skeletal sites, with no increase in the incidence of proliferative, benign or malignant neoplastic bone changes in the 10 μ g/kg/day dose group. Although the application of these findings to humans with osteoporosis needs to be carefully considered, the noncarcinogenic dose in rats occurred at systemic exposure 4.6-times that in humans at the proposed clinical dose. Recombinant human PTH is currently the only anabolic agent being investigated for the treatment of osteoporosis with a demonstrated noncarcinogenic dose in a 2-year rat bioassay.

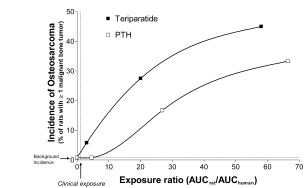


FIGURE 8.—Comparison of the osteosarcoma dose-response curves in Fischer 344 rats following treatment with PTH or teriparatide for 2 years. AUC = Area under the curve as estimate of exposure; PTH = recombinant human PTH 1–84.

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Abaloparotide and teriparatide enhance mandibular growth in adolescent rats with site-specific and mechano-related effects

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Summary

Objective: Teriparatide (TPD) and abaloparotide (ABL) are two osteoanabolic drugs targeting parathyroid hormone (PTH)1R signalling. This study aimed to investigate the effects of TPD and ABL on the adolescent mandibular growth.

Method: In total, 70 4-week-old male Sprague-Dawley rats were randomly divided into 14 groups, treated with intermittent TPD or ABL at various doses, accompanied by mandibular advancement (MA) or not. 3D printing was used to fabricate an innovative splint for MA. After a 4-week treatment, morphological measurement, histological and immunohistochemical analysis were performed. Mandibular condylar chondrocytes (MCCs) were treated with TPD or ABL, followed by CCK-8 assay, alcin blue staining, real time-PCR and immunofluorescent staining.

Result: In vivo, TPD or ABL alone increased the condylar length and cartilage thickness, with up-regulated SOX9 and COL II, whilst down-regulated COL X; however, when combined with MA, the promotive effects were attenuated. TPD or ABL alone increased the mandibular body height and mandibular angle width, whilst increased the mandibular body length and alveolar bone width when combined with MA. In vitro, TPD or ABL enhanced the MCC proliferation, glycosaminoglycan synthesis, COL II and SOX9 expression, whilst down-regulated COL X, Ihh and PTHrP expression.

Conclusion: Both ABL and TPD enhance mandibular growth in adolescent rats with site-specific and mechano-related effects, including promoting chondrogenesis at the condylar cartilage and promoting bone apposition at other mechano-responsive sites. They behave as promising drugs for mandibular growth modification, and in general ABL seems more potent than TPD in this context.

Introduction

Class II malocclusion is a common dentofacial malocclusion, and among the class II patients seeking for orthognathic surgery treatment, as high as 89% are inflicted with mandibular retrusion (1, 2). Mandibular hypoplasia (MH) not only affects the patient's facial appearance and occlusal function, but also increases the risk of obstructive sleep apnea-hypopnea syndrome (OSAHS) (3), which grievously harms health and growth. Orthognathic surgery can ameliorate severe MH in adulthood; however, before that, the patient has to suffer from its detriment both physically and psychologically. Therefore, if MH could be improved, more or less, during the growing period, it would be considerably beneficial for the patient's occlusal function, respiratory capacity, facial appearance and mental health, with the difficulty of surgery reduced as well. For adolescent patients with MH with growth potential, functional appliances are recommended by some orthodontists (4). However, a wide range of clinical studies has shed light on the frustrating conclusion that functional orthopaedic treatment alone has only quite a limited effect on promoting real skeletal mandibular growth (5, 6).

The growth sites of the mandible involve the condyle, the posterior border of the mandibular body, the sigmoid notch, and the alveolar crest (7), among which, intramembranous ossification of the mandibular main body and endochondral ossification of the condyle act as two major growth patterns (8, 9). The condyle for particular plays an essential role in augmenting mandibular length and height (10). During endochondral ossification of the mandibular condylar cartilages, the differentiation of osteochondral progenitors into chondrocytes is regulated by SHY-related high mobility group box gene 9 (SOX9) (11). SOX9 directly triggers the expression of type II collagen (COL II) (12), which is produced by chondrocytes as a major collagen component of the cartilage matrix (13). Another cartilage matrix component, type X collagen (COL X), is synthesized by hypertrophic chondrocytes and its expression indicates termination of chondrogenesis with subsequent initiation of endochondral ossification (14, 15). The growth and endochondral ossification of the mandibular condyle in adolescence could be modulated by a variety of external signals, including mechanical stresses (16) and biomolecules such as insulin-like growth factor-1 (IGF-1) (17), parathyroid hormone (PTH) (18), parathyroid hormone

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related protein (PTHrP) (19), Indian hedgehog (Ihh) (20), etc. Since mechanical stimulation with functional appliances alone could not greatly increase mandibular growth, it is attractive to attempt the strategy of combining functional orthopaedic treatment with medications that have pro-chondrogenic and/or osteoanabolic potentials.

The PTH is known as a regulator of calcium homeostasis (21). Its paradoxical dual effects on bone metabolism have been widely studied, that is, continuous administration causes catabolic effects whilst intermittent dosing is anabolic (22). Based on the role of PTH in promoting bone anabolism, teriparatide (TPD), a peptide that is structurally identical to human PTH (1–34), has been approved by FDA as the first osteoanabolic drug for treating osteoporosis (23). On the contrary, PTHrP which is originally found in human malignant hypercalcemia, has similar osteoanabolic effects and shares the same type 1 PTH/PTHrP receptor (PTH1R) with PTH (24). However, the short half-life of PTHrP makes it easy to degrade and therefore difficult to take effect in systemic administration (25). In order to exploit the function of PTHrP and overcome its unsuitability, abaloparotide (ABL), a synthetic analogue of PTHrP (1–134) has been developed and also approved for the treatment of osteoporosis (26). Although TPD [or PTH (1–34)] and ABL seem to have similar osteoanabolic effects, their osteolytic potential may discriminate due to different relative affinities to the RG and R0 conformations of PTH1R, resulting in less bone-resorptive effects of ABL compared to TPD (24). In gross, ABL is believed to overcome the disadvantage of easy decomposition of PTHrP (1–34) and attenuate the bone metabolic window loss caused by TPD.

Regarding cartilage, both PTH and PTHrP exert regulatory effects on mandibular condylar chondrocytes. PTH enhances the chondrocyte proliferation and increases the expression of COL II with a concomitant decrease of COL X and runt-related transcription factor 2 (RUNX2), indicating the promotion of chondrogenesis and suppression of terminal chondrocyte differentiation (18, 27). PTHrP plays an essential role in the prenatal and postnatal growth of the condyle (8, 28). The PTHrP-deficient mice show deformities of the ramus and mandibular body, indicating that PTHrP disruption may impair both intramembranous and endochondral ossogenesis of the mandible (8). PTHrP may regulate the proliferation, chondrogenic and osteogenic differentiation of chondrocytes by up-regulating the expression of SOX9 and COL II, inhibiting the expression of RUNX2, COL X, and ALP (29–31). In the long bones of limbs, which are predominant with endochondral ossification, the negative feedback loop formed with PTHrP and Ihh has long been recognized to regulate the lengthening of the growth plate. Notably, recent finding revealed skeletal stem cells among PTHrP-positive chondrocytes within the resting zone of the postnatal growth plate (32). PTHrP might also interplay with Ihh to maintain homeostasis of the mandibular condylar cartilage (33). Both *in vivo* and *in vitro* studies have demonstrated force-induced expressions of Ihh and PTHrP in the mandibular condylar cartilage, suggesting them as mechanotransduction mediators (29, 34).

In brief, PTH and PTHrP are both involved in the regulation of mandibular growth, and their derivative osteoanabolic drugs, TPD and ABL, respectively, may well have the potential to promote the growth of the mandible in adolescents. Thus, this study was aimed to investigate the effects

of intermittent TPD or ABL on mandibular growth, with focus on the mandibular condylar cartilage, and furthermore, whether there were additive or synergistic effects when the drugs were combined with mandibular advancement (MA) mechanical stimulation.

Materials and methods

Animal model and experimental design

The study was approved by the Animal Care and Ethics Committee of the State Key Laboratory of Oral Disease, West China School of Stomatology, Sichuan University, China (No. WCHSIRB-D-2017-259). All the experimental procedures followed the guidelines for animal experimentation of the university/the Animal Research N3CRs guidelines for Reporting of In vivo Experiments (ARRIVE) guidelines.

In total, 70 male Sprague-Dawley rats (4-week-old, body weight 105 g $\pm$ 10 g) were randomly allocated into 14 groups ($n = 5$ in each), including one control group, one MA group, 6 drug groups and 6 drug + MA groups. The control group received a daily subcutaneous injection of normal saline (NS), whilst the MA group received a functional appliance for MA with a daily subcutaneous injection of NS. The 6 drug groups include the high-dose TPD (TPD^{high}, 8 mg/kg), mid-dose TPD (TPD^{mid}, 800 μ g/kg), low-dose TPD (TPD^{low}, 80 μ g/kg), high-dose ABL (ABL^{high}, 8 mg/kg), mid-dose ABL (ABL^{mid}, 800 μ g/kg), and low-dose ABL (ABL^{low}, 80 μ g/kg) group, which received a daily subcutaneous injection of TPD or ABL at the corresponding dose level. The 6 drug + MA groups include the TPD^{high} + MA, TPD^{mid} + MA, TPD^{low} + MA, ABL^{high} + MA, ABL^{mid} + MA, and ABL^{low} + MA group, which received both MA and daily subcutaneous injections of TPD or ABL at the corresponding dose level (Supplementary Figure 1). All the subcutaneous injections were given at the back of the neck. All the rats were housed in a constant temperature of 25°C, a 12-hour light/12-hour dark environment, and fed with a standard soft diet and water *ad libitum*.

Design and fabrication of the mandibular advancement appliance

It is noteworthy that the appliance for MA was designed via digital design and 3D printing to optimize the accuracy of mandibular repositioning in the rat model. Firstly, the digital model of an inclined bite splint was designed on Tinkercad software (<https://www.tinkercad.com>), and then the solid male mold was made through 3D printing. The bite splint is an isosceles trapezoid with a length of 4 mm at the top, 8 mm at the bottom and a height of 7 mm. From side view, it is a straight-angle trapezoid with a thickness of 2 mm (Figure 1A and 1B). Innovatively, a smooth groove was designed on the occlusal surface of the guide plate to prevent mandibular deviation during MA (Figure 1A arrow). The male mold was placed in the heated liquid silicone gel and then solidified to obtain the template of the bite splint (Figure 1C). Then, the resin was filled in the template and bonded on rat upper incisors under anaesthesia with 70 mg/kg ketamine (Ketavet 100, Intervet Productions Srl, Aprilia, Italy) and 5 mg/kg xylazine (Sedaxylan, Eurovet Animals Health B.V., Bladel, The Netherlands), making sure that the lingual surface of the upper incisor was in contact with the back wall of the template without left-right or forward-backward deflection

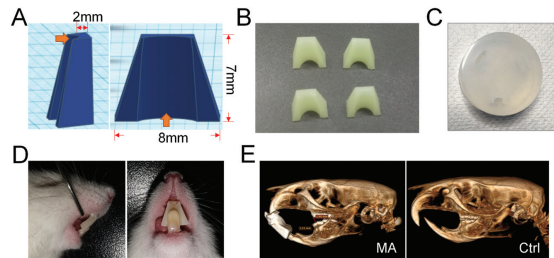


Figure 1. The design, fabrication and wearing of the innovative mandibular advancement (MA) appliance. (A) The digital design of the inclined bite splint, with a central groove (orange arrow) prevents mandibular deviation during advancement. (B) The 3D-printed male model. (C) The template is made of silica gel. (D) The frontal and lateral views of the bite splint in the rat mouth. (E) Three-dimensional reconstruction of cone beam computed tomography showed the mandible was displaced forward and downward after the use of the mandibular advancement appliance.

(Figure 1D). Cone beam computed tomography imaging confirmed that the mandible was displaced forwards and downwards after the use of the MA appliance (Figure 1E).

Morphological measurements

The rats in each group were euthanized using excessive anaesthetic injection after a 4-week treatment. The rat mandibles were dissected mid-sagittal and divided into left and right halves without muscle tissue attached to the bone surface. The mandibular morphology was measured referring to the reported method (35). Briefly, pictures of the mandibles were taken using a digital single lens reflex camera (Nikon D700, Japan), which was mounted at a 90° angle and at a fixed distance from the samples. The mandibular length and height in these photos were measured with Image-Pro plus (version 6.0, Media Cybernetics, Rockville, MD, USA), whilst the width of the mandible was measured with vernier callipers. Measurement items included the mandibular condylar length, the mandibular body length, the mandibular body height, the mandibular incisal alveolar bone width, and the mandibular angle width, with the length and height standardized to the whole-body length. The selected landmarks and measurement items were shown in Table 1 and Figure 2A.

Histology and immunohistochemistry assays

The condyles were harvested after digital pictures were taken. The specimens of condyles were fixed in 4% paraformaldehyde for 24 hours and subsequently decalcified in 10% ethylene diamine tetraacetic (EDTA) for 8 weeks at 37°C. Subsequently, gradient-dehydrated, dipping wax, and paraffin-embedding were performed before sectioning condyles at 5 µm thickness to obtain serial sagittal slices.

These slices were stained with safranin O/fast green, or haematoxylin and eosin (HE), for cartilage demarcation and morphological measurement. The condylar cartilage was divided into four layers that is, the fibrous, proliferative, chondrogenic, and hypertrophic layer (10, 36) (Figure 3A). Since the posterior region of the condylar cartilage was reported to be the most sensitive to MA stimulation (11, 36),

the posterior region of the condylar cartilage in front of the articular disc attachment was selected as the region of interest (ROI). Images of the samples were obtained using a light microscope (Leica DM 2500, Wetzlar, Germany), and the thickness of each layer and the whole cartilage were measured using Image-Pro plus6.0. Specifically, a straight line perpendicular to the surface of condylar cartilage was made on the HE-stained image, which ran through the fibrous, proliferative, chondrogenic, and hypertrophic layers. The length of the line in each layer was calculated by the linear measurement tool of Image-Pro plus 6.0 software as the thickness of this layer. The total thickness was the sum of the thickness of each layer (Figure 3A).

For the immunohistochemistry assay, the slices were incubated with primary antibodies (1:200) against SOX9 (ab185966; rabbit, Abcam, Cambridge, MA, USA), COL II (ab34712; rabbit, Abcam, Cambridge, MA, USA), and COL X (ab260040; rabbit, Abcam, Cambridge, MA, USA) overnight at 4°C. Incubation with PBS instead of the primary antibody was used as a negative control. Following rinsing with PBS for three times, the samples were incubated with HRP-labelled goat anti-rabbit IgG polymer at 37°C for 20 minutes according to the instruction (PV-9001, ZSQC-Bio, Beijing, China). After rinsing with PBS for three times, freshly prepared DAB chromogen was applied and incubated for 5 minutes at room temperature (ZLI-9018, ZSQC-Bio, Beijing, China).

Similar to HE-stained sections, the ROI for immunohistochemical semi-quantitative analysis was also delineated in the posterior region of the condylar cartilage before the articular disc attachment point. Mean optical density (MOD) in the ROI of each sample, that is, the ratio of integrated optical density (IOD) to the ROI area, was calculated with Image-Pro plus6.0. Given the different expression distribution of the three target proteins, the ROI of SOX9 was in the proliferative and chondrogenic layers, COL II in the chondrogenic and hypertrophic layers, and COL X in the hypertrophic layer. Each sample was measured three times under ×400 magnification, and the average value was taken as the result of each condyle sample.

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Table 1. Definition of landmarks and measurement items.

Landmarks	Definition
L1	The most anterior point of the lingual alveolar bone of lower anterior teeth
Mf	The midpoint of the mandibular foramen
A	The most anterior point of the condylar
B	The most posterior point of the condyle
C	The midpoint between A and B
Co	The intersection point of the extension line of Mf-C and the outer contour of the condyle
Me	The most posterior point where the digastric muscle attaches to the lower margin of the mandible
K	The most prominent point of the alveolar bone of the lower anterior teeth
K'	The projection of point K on the inner side of the mandible
M	The most posterior point of the angular process
M'	The projection of point M on the inner side of the mandible
MP plane	The plane tangent to the mandibular lower margin through Me
Measurements	
Length	
Co-Mf (mm)	The condylar length
Mf-L1 (mm)	The mandibular body length
Height	
Mf-MP (mm)	The distance from Mf to MP plane represents the mandibular body height
Width	
K-K' (mm)	The mandibular incisal alveolar bone width
M-M' (mm)	The mandibular angle width

Mandibular condylar chondrocyte isolation and culture

Mandibular condylar cartilage was dissected from SD rats within 24 hours of birth. In brief, the mandibular condyles of new-born rats were harvested in penicillin-streptomycin (PS, R20016, Hyclone, Wuhan, China) and PBS solution. Subsequently, the condylar cartilages were dissected with microforceps after removing the soft tissue on their surface under a stereomicroscope. Then, cartilages were cut into small pieces followed by sequential digestion with trypsin (25200072, 0.25%, Gibco, USA) for 20 minutes and type II collagenase (YZ-1148090, 0.2%, Solarbio, Beijing, China) for 1 hour at 37°C. After resuspension and filtration, the chondrocytes were dispersed into single cells and cultured in DMEM/F12 supplemented with 10% FBS (10100147, Gibco, USA), and 1% PS under a humidified atmosphere of 5% CO₂ at 37°C. Upon confluency, the primary cells were further subcultured to passage 1 (P1) for *in vitro* experiments. Alcian blue and safranin O staining were used to identify the MCCs.

Cell counting Kit-8 assay

The cell counting Kit-8 (CCK-8) assay (C0037, Beyotime, Shanghai, China) was carried out to investigate the effects of TPTD and ABL on MCCs proliferation. MCCs (5 × 10³ cells/well) were seeded in 96-well cell plates and treated with 100 nmol/ml TPTD or ABL, and the media were changed every

24 hours for 3 days. Standard curves were constructed by measuring the optical density (OD) value at 450 nm.

Alcian blue staining

P1 MCCs (5 × 10⁴ cells/well) were seeded in 24-well cell plates and treated with 100 nmol/ml PBS, TPTD, or ABL for 48 hours. The glycosaminoglycan (GAG) content of MCCs was identified using a standard alcian blue staining kit (A8140, Solarbio, Beijing, China). The cells were washed three times with PBS, fixed with 4% paraformaldehyde at room temperature for 20 minutes, again washed three times with PBS (3 minutes each time), stained with alcian staining solution for 30 minutes, and washed three times with PBS. The images were acquired using a stereomicroscope, with consistent photographic parameters.

Total RNA extraction and quantitative real-time PCR analysis

The P1 MCCs (15 × 10⁴ cells/well) were seeded in six-well cell plates and incubated with 100 nmol/ml PBS, TPTD or ABL for 24 hours. Total RNA was isolated from MCCs using RNA-Quick purification kit (ES-RN001, ES Science, Shanghai, China) according to manufacturer's instructions, and PrimeScript RT reagent kit (RR047A, TaKaRa, Japan) was used for reversed transcription of RNA for cDNA. Quantitative real-time PCR was performed using an SYBER Green qPCR kit (B21703, Bimake, USA), and carried on in StepOne Real-Time PCR System (Applied Biosystems). Glyceraldehyde 3-phosphate dehydrogenase (GADPH) was used as an internal control gene to normalize the expression of other genes, including *Col2a1*, *Col10a1*, *Sox9*, *Ihh*, *Pthlh*, *Pth1r*. The results were calculated using the 2^{-ΔΔCT} method and presented as fold increases relative to the control group. The primer sequences are described in the Supplementary Table 1.

Immunofluorescence staining

P1 MCCs (3 × 10⁴ cells/well) were seeded on 48-well plates, and incubated with 100 nmol/ml PBS, TPTD or ABL for 24 hours. Then MCCs were washed by PBS and fixed in 4% paraformaldehyde for 30 minutes, permeabilized in 0.1% Triton X-100 at room temperature for 10 minutes (for COL II, COL X and Ihh) or 0.5% Triton X-100 on ice for 20 minutes (for SOX9 and PTHP) and then blocked with 3% BSA at room temperature for 30 minutes, before incubated overnight with rabbit anti-rat COL II (1:200, ab34712, rabbit, Abcam, Cambridge, MA, USA), -COL X (1:200, ab260040, rabbit, Abcam, Cambridge, MA, USA), -SOX9 (1:200, ab185966, rabbit, Abcam, Cambridge, MA, USA), -Ihh (1:200, DF7688; rabbit, Affinity, China), -PTHP (1:200, DF6532, rabbit, Affinity, China), and -PTHR (1:200, DF6589, rabbit, Affinity, China) antibody at 4°C. The cells were incubated with goat anti-rabbit secondary antibody (1:200, A0516, goat, Solarbio, Beijing, China) and phalloidin (1:400, C2201S, Solarbio, Beijing, China) for 1 hour, then washed twice with PBST, and incubated with DAPI for 10 minutes. The MCCs were finally washed with PBS 3 times and analyzed with Leica fluorescence microscope (Leica, Wetzlar, Germany).

Statistical analysis

Data analysis was performed using SPSS version 25.0 (Statistical Package for the Social Sciences, SPSS, Chicago, IL, USA). When

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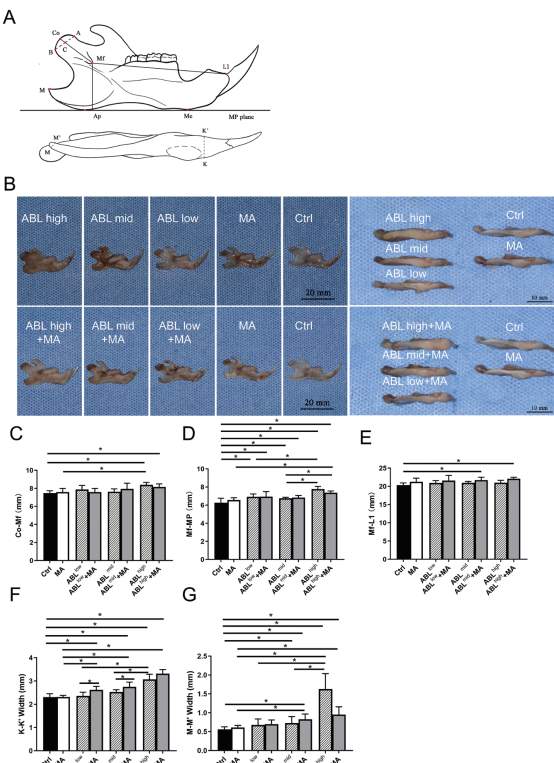


Figure 2. Morphological measurement and analysis of the mandible in ABL-treated groups. (A) Illustration of landmarks and measurement items (see Table 1 for definitions). (B) Photographs of the mandibles. (C) The condylar length (Co-Mf). (D) The mandibular body length (Mf-L1). (E) The mandibular body height (Mf-MP). (F) The mandibular lower anterior alveolar bone width (K-K'). (G) The mandibular angle width (M-M').

the variances were equal in the Levene's test for homogeneity of variance, a one-way analysis of variance (ANOVA) followed by post hoc least significant difference (LSD) test was used;

otherwise, Welch's test followed by Dunnett's T3 test was applied. Statistical significance was set as $P < 0.05$. All quantitative data were presented as mean ± standard deviation (SD).

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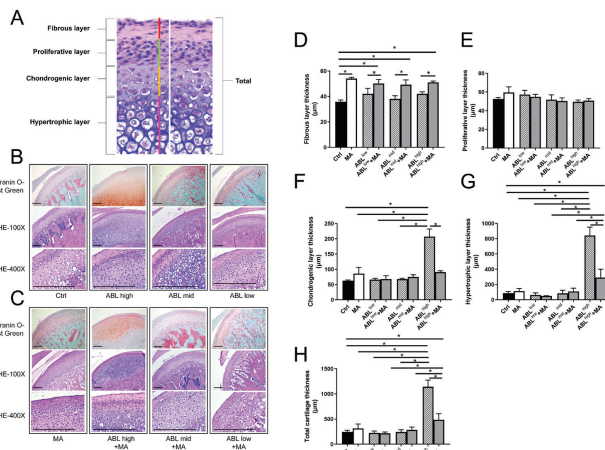


Figure 3. Histological staining and thickness measurement of the mandibular condylar cartilage in ABL-treated groups. (A) Schematic diagram of the various layers of condylar cartilage. The colour lines indicate the measured length of each layer, and the white line indicates the total cartilage thickness. (B, C) HE and Safranin O staining. Thickness analysis of the fibrous layer (D), proliferative layer (E), chondrogenic layer (F), hypertrophic layer (G), and total cartilage (H). (*Significant difference of $P < 0.05$, scale bar = 200 µm).

Results

ABL and TPTD had synergistic effects with MA mechanical loading to enhance mandibular growth

Morphological measurement showed that the condylar length (Co-Mf) in the TPTD^{high} and ABL^{high} groups were significantly greater than that in the control group, whilst mid- or low-dose TPTD or ABL had no effects on the condylar growth (Figure 2C, Supplementary Figure 2B).

TPTD and ABL also promoted growth of the mandibular body, including its height and length. All the doses of ABL, with or without MA, significantly increased the mandibular body height (Mf-MP) (Figure 2D). In contrast, TPTD could function independently only at high dose, however effective at mid- and low-doses only when combined with MA (Supplementary Figure 2C). Moreover, high-dose ABL was more effective than high-dose TPTD in mandibular body height increment (Supplementary Figure 3C). Interestingly, although TPTD, ABL, or MA alone had no significant effects on Mf-L1 in the TPTD^{high} and ABL^{high} groups, when high- or mid-dose TPTD or ABL was combined with MA, a significant increase in Mf-L1 was observed (Figure 2E, Supplementary Figure 2D), indicating a synergistic effect of the drugs and MA mechanical loading on the growth of the mandibular body.

Regarding width, TPTD or MA alone did not elicit significant changes in the lower anterior alveolar bone width (K-K'). However, when TPTD was combined with MA, regardless of the drug dose, the alveolar bone width was significantly increased compared to that of the control or MA group. In contrast to TPTD, high- or mid-dose ABL alone significantly increased the lower anterior alveolar bone width, and when combined with MA, the increase became significantly larger compared to the ABL alone groups (Figure 2F, Supplementary Figure 2E). In addition, when the two drugs were compared, the increase of alveolar bone width in the ABL^{high} group was significantly larger than that in the TPTD^{high} group (Supplementary Figure 3D). Furthermore, high- or mid-dose TPTD or ABL, whether combined with MA or not, significantly increased the mandibular angle width compared with the control (Figure 2G, Supplementary Figure 2F).

Taken together, both ABL and TPTD enhanced the mandibular growth with effects varying with different site. Of note, synergistic effects of the drugs and MA mechanical loading were shown at the mandibular body and the alveolar bone. In general, ABL demonstrated bone-augmenting effects more powerful than TPTD in the present context.

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ABL increased the condylar cartilage thickness even more than TPTD, which was attenuated by MA. HE staining showed that MA, but not TPTD or ABL, significantly increased thickness of the fibrous layer, indicating that thickening of this layer was an adaptive response to the mechanical stimulation of MA (Figure 3D, Supplementary Figure 4C). As for the proliferative layer, only the TPTD^{high} group had a significant decrease in the proliferative layer thickness, while the other experimental groups did not differ with the control group (Figure 3E, Supplementary Figure 4D). Moreover, the chondrogenic layer, hypertrophic layer and the total cartilage were all significantly increased in thickness only when high-dose TPTD or ABL was used (Figure 3F–H, Supplementary Figure 4E–G), and the increase was significant larger in the ABL^{high} group than TPTD^{high} group (Supplementary Figure 3F). However, when high-dose TPTD or ABL was combined with MA, thickness of the chondrogenic layer, hypertrophic layer and total cartilage was lower than that of the drug alone groups (still higher than the control group) (Figure 3F–H, Supplementary Figure 4E–G).

Collectively, both high-dose ABL or TPTD alone could increase the condylar cartilage thickness, with ABL more effective than TPTD; however, the combination of the drugs with MA attenuated the promotive effects.

ABL and TPTD promoted chondrogenesis and inhibited hypertrophy of the condylar cartilage, which was partially counteracted by MA

Immunohistochemistry staining showed that the expression of SOX9 was significantly up-regulated by MA, high-dose TPTD, or ABL of any doses, and the combination of MA and drugs had an additive effect (Figure 4A and 4B, Supplementary Figure 5A and 5B). However, the COL II expression was significantly up-regulated only in the TPTD^{high} and ABL^{high} groups, but not in the TPTD^{mid} + MA or ABL^{mid} + MA groups (Figure 4C and 4D, Supplementary Figure 5C and 5D). In addition, the expression of COL X was significantly attenuated in the TPTD^{high}, TPTD^{mid} and ABL^{mid} groups, but not in the corresponding drug + MA groups (Figure 4E and 4F, Supplementary Figure 5E and 5F).

Comparing TPTD and ABL, high-dose ABL exerted stronger up-regulation of SOX9 than high-dose TPTD with or without MA; nevertheless, they showed no significant difference in the regulation for COL II or COL X expression (Supplementary Figure 3H and 3I).

In summary, ABL and TPTD promoted chondrogenesis and inhibited hypertrophy of the mandibular condylar cartilage, with ABL being more effective than TPTD to up-regulating the SOX9 expression; however, these effects were partially counteracted by MA in terms of the COL II and COL X expression.

ABL and TPTD enhanced proliferation and chondrogenic differentiation whilst inhibited hypertrophy of MCCs *in vitro*

CKK-8 assay showed that both ABL and TPTD significantly enhanced the cell viability of MCCs compared with the control at 48 hours and 72 hours, with no significant difference between the two drugs (Figure 5B). After incubated with ABL or TPTD for 24 hours, real-time PCR (Figure 5C–H) showed up-regulated *Sox9* and *Col2a1*, down-regulated *Col10a1*, *Ihh*

and *Pth1r*, whilst unchanged *Pthlh*. When comparing the two drugs, ABL was stronger at up-regulating *Sox9*, while TPTD was better at down-regulating *Ihh*. In addition, alcian blue staining showed significantly increased GAG in the experimental groups compared with the control at 48 hours (Figure 5I). The immunofluorescence staining results (Supplementary Figure 6) were consistent with the PCR.

Together, ABL and TPTD enhanced proliferation and chondrogenic differentiation, whilst inhibited hypertrophy of MCCs *in vitro*, with ABL more effective than TPTD to up-regulating the SOX9 expression.

Discussion

For a long, functional appliance treatment seems to be the only clinical approach to promote mandibular growth in adolescence, with limited real skeletal effects. It is of great importance to pursue other biotherapies for mandibular growth modification. Many biomolecules and physical stimulations have been shown to enhance the growth of the mandibular condylar cartilage (37–39). In the present study, we observed and compared the effects of the two FDA-approved osteoanabolic drugs, TPTD and ABL, on promoting mandibular growth in adolescent rats, combined with or without MA mechanical loading.

The various effects of the two drugs on different sites of the mandible were most interesting (Figure 6). Overall, their effects were similar, and ABL seemed to be more potent than TPTD. ABL alone increased the condylar length, the mandibular body height and the mandibular angle width, which could not be further enhanced by MA; ABL alone increased the anterior alveolar bone width, which could be further enhanced by MA; ABL could not increase the mandibular body length unless combined with MA. Thus, the MA splint seemed to elevate the mechanical stress level only at the anterior section of the mandible, which amplified the effects of ABL. Taken together, the anabolic modelling effects of ABL and TPTD on the mandible are associated with mechanical stress, both from physiologic activities and the MA splint (Figure 6).

Notably, both ABL and TPTD could increase the mandibular anterior alveolar bone width. In clinical, function appliances may labialize the lower anterior teeth, which may result in dehiscence (44, 45). Thus, augmentation of the anterior alveolar bone width with the drugs can help lower periodontal risks in MA. On the other hand, the increment of mandibular body height by ABL or TPTD (Figure 2B, Supplementary Figure 2A) may lead to counter clockwise rotation of the mandible, which has potential clinical implication for growth modification of high-angle adolescent patients with low-posterior facial height.

The mandibular angle width was increased by both TPTD and ABL alone, whilst TPTD had no effects on alveolar bone width unless combined with MA. This difference could be explained with differential mechanical microenvironment between the two sites. The mandibular angle is subject to strong and frequent physiological mechanical stimulation by the masticatory muscles; in contrast, the mandibular anterior alveolar bone undergoes much less physiological mechanical loading, partially due to excessive overjet of the rat dentition, which may be greatly enhanced by the MA splint. In fact, the combinative effects of mechanical stress and PTH or PTHrP have been well-established in osteoporosis studies (46–48).

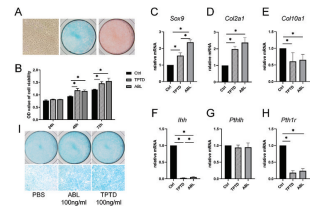


Figure 5. The proliferation and mRNA expression of the mandibular condylar chondrocytes (MCCs). (A) Morphology of the primary MCCs and identification with alcian blue and Safranin O staining. (B) Cell viability of the MCCs at 24, 48, and 72 hours. mRNA expressions of *Sox9* (C), *Col2a1* (D), *Col10a1* (E), *Ihh* (F), *Pthlh* (G) and *Pth1r* (H) of the mandibular condylar chondrocytes after 24-hour TPTD or ABL treatment. (I) Alcian blue staining of the MCCs treated with TPTD or ABL for 48 hours. (*Significant difference of $P < 0.05$).

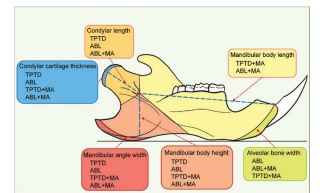


Figure 6. Summary for the site-specific growth promotive effects of intermittent TPTD or ABL at various sites of the mandible in adolescent rats, accompanied with MA mechanical loading or not.

Regarding molecular mechanisms, the immunohistochemical results showed that the expression of SOX9 was up-regulated by TPTD, ABL, or MA alone, among which ABL exerted the strongest effect. When TPTD or ABL was combined with MA, the SOX9 up-regulation was further enhanced. Both TPTD and ABL up-regulated COL II, whilst down-regulated COL X, suggesting that TPTD and ABL could promote chondrogenesis and inhibit chondrocyte premature hypertrophy.

The findings *in vitro* were consistent with *in vivo*, that is, CKK-8, real-time PCR and immunofluorescence confirmed that ABL and TPTD could promote proliferation and chondrogenesis, while inhibit hypertrophy of MCCs. Moreover, the Ihh-PTHrP negative feedback loop has been reported to maintain homeostasis of the condylar cartilage (50, 51). In the present study, exogenous TPTD and ABL reduced the expression of *Ihh*, which could promote hypertrophy of MCCs; the endogenous PTHrP expression was not changed, whilst the PTHrP expression of MCCs was suppressed by either TPTD or ABL. Their effects on PTHrP were consistent with previous studies (52–54), and the down-regulation of PTHrP might attenuate the effects of exogenous TPTD or ABL.

Although our study demonstrated that ABL and TPTD are efficacious in promoting mandibular growth, there are still some limitations to this study. Firstly, we used subcutaneous injection according to the clinical recommendations of ABL and TPTD, leading to the inevitable systemic effects. Secondly, this study only focused on the effects of ABL and TPTD on the mandible, and their impacts on other craniofacial structures should be taken into account in future studies. Additionally, since PTHrP is an autocrine/paracrine factor, this study was performed in a male rat model out of concern that females might be more susceptible to the pubertal sex hormones. Admittedly, gender difference has been reported in the rapid developmental changes of rat condylar cartilage and subchondral bone (55). Therefore, further investigation into the female rat models would be necessary.

Conclusions

Both ABL and TPTD enhance mandibular growth in adolescent rats, which is site-specific and mechano-related (Figure 6). At the endochondral ossification site, they increase the mandibular condylar cartilage thickness via propelling chondrogenesis. At the intramembranous ossification sites, they promote the mandibular growth most evidently at the mandibular angle, which is mechanically hyperactive. When combined with MA, they increase the mandibular body length and anterior alveolar bone width. In general, ABL seems more potent than teriparatide for these effects.

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Conflict of interest

None to declare.

Data availability

The authors confirm that the data supporting the findings of this study are available within the article.

Supplementary material

Supplementary material is available at *European Journal of Orthodontics* online.

Supplementary Figure 1 The study flow chart. Briefly, after 7-day acclimation to the laboratory environment, seventy 4-week-old SD rats were randomly divided into 14 groups, and the experimental groups were treated with TPTD or ABL alone at different doses, MA alone, or combination of the drug and MA.

Supplementary Figure 2 Morphological measurement and analysis of the mandible in TPTD-treated groups. (A) Photographs of the mandibles. (B) The condylar length (Co-M). (C) The mandibular body height (Mi-MP). (D) The mandibular body length (M-L1). (E) The mandibular lower anterior alveolar bone width (K-K'). (F) The mandibular angle width (M-M').

Supplementary Figure 3 Comparison between the high-dose TPTD and the high-dose ABL with or without MA. (A, B) Comparison of the mandibular length. (C) Comparison of the mandibular height. (D, E) Comparison of the mandibular

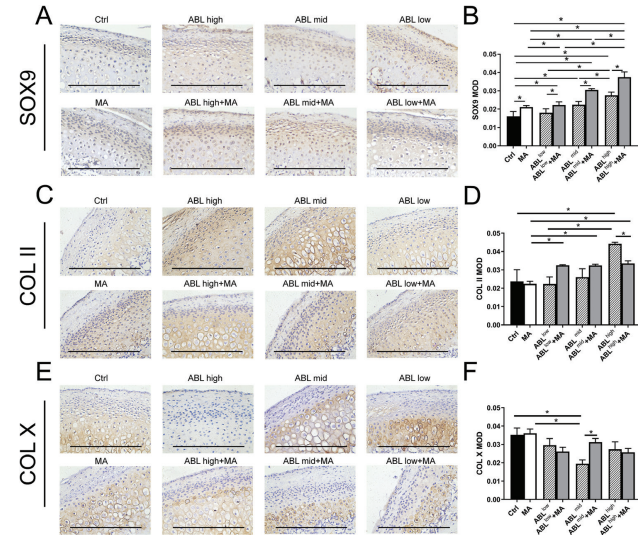


Figure 4. Immunohistochemical staining and semi-quantitative analysis showed that high-dose ABL significantly enhanced the expression of SOX9 (A, B) in the proliferative and chondrogenic layers and COL II (C, D) in the chondrogenic and hypertrophic layers, while the expression of COL X (E, F) was down-regulated. The synergistic effect of ABL with mandibular advancement was only found in their regulation of SOX9. (*Significant difference of $P < 0.05$, scale bar = 200 μ m).

The mandible is a composite bone formed via both endochondral ossification at the mandibular condylar cartilage and intramembranous ossification at other sites (40). As an essential growth site to increase the mandibular length and height, the condyle has been paid great attention to. The present study showed that TPTD or ABL alone significantly enhanced the condylar length. Accordingly, the total thickness of the condylar cartilage was significantly increased when high-dose TPTD or ABL was used, with ABL being the more effective. Notwithstanding, the MA splint did not significantly increase the cartilage thickness except that of the fibrous layer. The finding was largely similar to some previous studies (10), however inconsistent with others (41–43). Furthermore, the combination with MA even attenuated the effects of the two drugs on cartilage thickening. The MA splint used in the present study was made via digital design and 3D printing, which should be more accurate than the previous counterparts. It seemed that the effects of such MA splints on the rat condylar cartilage thickness might be transient and unpredictable.

A wide range of evidence indicates that PTH and PTHrP can enhance chondrocyte proliferation (18, 19, 27, 36); however, the HE staining showed unchanged or even decreased thickness of proliferative layers in the present study, which was somewhat counterintuitive. Condylar cartilage is a secondary cartilage, in which the differentiated chondrocytes no longer exhibit division ability, which means that all chondroblasts and hypertrophic chondrocytes derive from the differentiation of the upper cells rather than their own replication (49). In addition, BrdU labelling also demonstrated cell migration and differentiation from the proliferative layer to the chondrogenic and hypertrophic layers in the mandibular condylar cartilage (29). Therefore, although the proliferative layer thickness showed no increase after 4-week TPTD or ABL treatment in this study, significant thickening of chondrogenic and hypertrophic layer suggested the temporarily enhanced expansion of the proliferative layer, which might have faded at the sacrifice time point (8 weeks of age). In addition, the CKK-8 results in the *in vitro* study also supports that TPTD and ABL promote the proliferation of the MCCs.

width. (F, G) Comparison of the thickness of each layer and the whole condylar cartilage. (H, I) Comparison of the semi-quantitative immunohistochemical analysis of SOX9, COL II, and COL X.

Supplementary Figure 4 Histological staining and thickness measurement of the mandibular condylar cartilage in TPTD-treated groups. (A, B) HE and Safranin O staining. Thickness analysis of the fibrous layer (C), proliferative layer (D), chondrogenic layer (E), hypertrophic layer (F), and total cartilage (G). (*Significant difference of $P < 0.05$, scale bar = 200 μ m).

Supplementary Figure 5 Immunohistochemical staining and semi-quantitative analysis showed that TPTD significantly enhanced the expression of SOX9 (A, B) in the proliferative and chondrogenic layers and COL II (C, D) in the chondrogenic and hypertrophic layers, while the expression of COL X (E, F) was down-regulated. (*Significant difference of $P < 0.05$, scale bar = 200 μ m).

Supplementary Figure 6 Immunofluorescence showed that the expression of SOX9 (A) and COL II (B) was up-regulated in mandibular condylar chondrocytes of the ABL and TPTD groups at 24 hours, while the expression of COL X (C), Ihh (D) and PTHrP (E) was down-regulated, and there was no significant difference in the expression of PTHrP (F), compared with the control group. (scale bar = 250 μ m).

Supplementary Figure 7 Body weight change during the experiment of TPTD-treated groups (A) and ABL-treated groups (B).

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